August 3, 2026

THE PLATONIC ELLIPTIC SURFACES

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Abstract. We construct certain special rational elliptic surfaces with four reduced singular fibres that are naturally attached to the platonic solids and belong to a group of surfaces complementing the well-known semi-stable Beauville surfaces. We describe the basic algebraic geometric properties of these surfaces, their associated Picard-Fuchs operators, integer sequences, Laurent polynomial representation and Apéry constants. On the arithmetic side, we discover the need for a refinement of the Dwork-expansion used to find the Euler factors of the fibres of these fibrations. This is explained by comparing the q-coordinates coming from the elliptic curve and the q-coordinate of the Picard-Fuchs equation and is also directly reflected in the shape of the monodromy matrices in the Frobenius basis.

Introduction

Starting from the three reflection groups of the platonic solids we study certain elliptic surfaces that are naturally attached to them. These surfaces have four reduced singular fibres, three of which are semi-stable. It turns out that these have many properties in common with the well-known Beauville cases of four semi-simple fibres, but also exhibit some interesting new features. We determine the Picard-Fuchs operators for these surfaces, which are Heun equations of a special type, from which one obtains a 2-term recursion relation for the expansion coefficients, forming integer sequences analogues to the (small) Apéry squence. We find that these surfaces and operators exhibit some unusual arithmetic properties, which is reflected in the monodromy of the operator and the Dwork-expansion that determines the Euler factors of the fibres of these elliptic surfaces.

This paper, which is partly expository in nature, is structured as follows.

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2020 *Mathematics Subject Classification.* 14J27.

Key words and phrases. platonic solids, elliptic surfaces, Apéry sequences.

In section 1 we describe how the reflection groups of the platonic solids give rise to an elliptic threefold, from which we obtain elliptic surfaces as sections, which we call *platonic elliptic surfaces*.

In section 2 we review some basic properties of elliptic surfaces in general and describe how these platonic elliptic surfaces fit in the classification and point out the differences with some other famous elliptic surfaces that have a clear relation to the platonic solids.

In section 3 we describe the Picard-Fuchs operators attached to (normalised) period integrals of our surfaces, which turn out to be special Heun operators with exponents $0, 0$ at the three finite singular points and of finite order at ∞.

In section 4 we discuss the integer sequences attached to these operators and present Laurent-polynomial expressions for all but two of these sequences, and note some congruence properties.

In section 5 overview some facts about the arithmetic of the elliptic curves appearing as fibres in our elliptic surfaces, using the method that goes back to Dwork. By comparing at a nodal fibre the Serre-Tate canonical q-parameter with the q-coordinate that emerges from the *critical integral scaling* of the Picard-Fuchs operator one finds the appearance of a logarithmic term in the limit Frobenius matrix, which is unlike what happens for the Beauville operators. We find that the monodromy matrices expressed in the Frobenius basis are affected by the logarithmic term in a corresponding way. This will be exlained in section 6.

In Appendix B some further data attached to our examples are collected.

Acknowledgement: We are grateful to Bradley Klee for calling our attention to his beautiful work [38] in which the platonic operators and some of the associated integer sequences are connected to the vibrational spectrum of molecules with platonic symmetry. Our desire to understand the relationship of the platonic solids, the elliptic surfaces and the special Heun operators resulted in this paper. Our attempts to follow his suggestion that these operators may be used to construct further fourth order Calabi-Yau operators have so far not met with succes. Further thanks to `Claude Fable`, which was of great help with verification and some extremely helpful hints in the later stage of this project.

We gratefully acknowledge the support by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project-ID 444845124 – TRR 326.

FIGURE 1. Roman dodecahedron

1. Platonic Solids and reflections groups

1.1. **Three reflection groups.** The five platonic solids appear ubiquitous in mathematics and popular culture. There are good reasons to believe they were known in prehistoric times [42], although the details of their discovery remain obscure. In the Scholium to book XIII of the Elements of Euclid, describing the construction of the platonic solids, ascribes the tetrahedron, cube and dodecahedron to the pythagoreans, octahedron and icosahedron to Thaetetus. We refer to [61] for a historical acccount. The story of the discovery of the dodecahedron by the Pythagorean Hippasos and his fate is told by Iamblicus [33], chapter XVIII. The precise use of the so-called *Roman dodecahedron* (Figure 1) found all over Europe is as yet unknown.

The basic geometry of the regular polyhedra belongs to the standard canon of elementary mathematics for which we refer to Coxeter [13], where among other things, the general theory of real reflection groups (Coxeter groups) and their Coxeter-Dynkin diagrams is developed.

The three dimensional reflection groups are labelled by *platonic triples* (p,q,r), with $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$, which encode the angles $\pi/p, \pi/q, \pi/r$ of the spherical triangle forming a fundamental domain (Figure 2) for the reflection group G. The cases of $(p,q,r)=(2,3,3),(2,3,4),(2,3,5)$ lead respectively to the tetrahedron, octahedron/hexahedron, icosahedron/dodecahedron, whereas the cases $(2,2,n)$ correspond to the dihedral groups.

The number of elements of the reflection group G attached to (p,q,r) is given by the uniform formula

$$|G| = \frac{4}{1/p+1/q+1/r-1}.$$

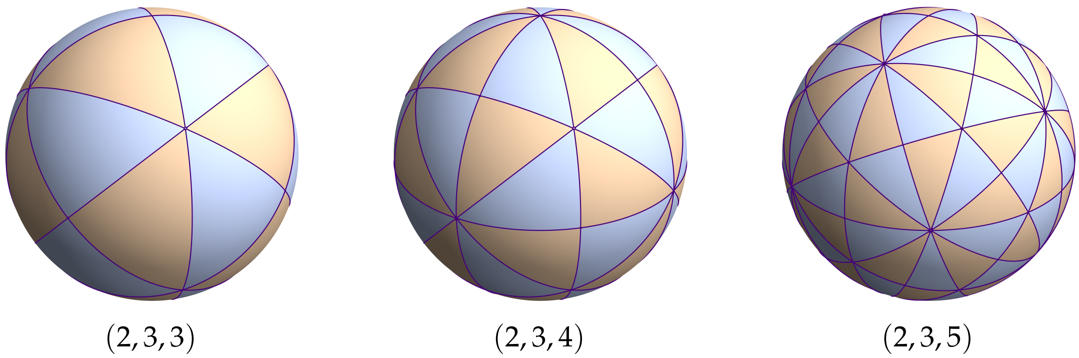


FIGURE 2. Spherical tessellations associated with the platonic triples.

This is also the number of elements of the associated *binary groups* $\mathbf{G}$, inverse image of rotation group $SG := SO(3) \cap G$ under the 2-1 covering $SU(2) \longrightarrow SO(3)$ that were studied by F. Klein in [37]. As recognised by DuVal [18], the corresponding quotient singularities $\mathbb{C}^2/\mathbf{G}$ are exactly those that can be resolved by a system of curves organised in a Dynkin graph of type ADE. Thus the reflection groups of tetrahedron, octahedron and icosahedron are related in this way to Dynkin diagrams E_6, E_7 and E_8, where the arm-lengths to the central node coincide with the platonic triple. This relation between two basic *trialities* in the sense of Arnol'd [4] was recently reconsidered in [14]. The ADE list also coincides with the list of simple singularities in the sense of Arnol'd, [3]. The ADE-graphs and their associated higher dimensional reflection groups describe the root systems of the simply laced Lie-algebras bearing the same names. For a more information highlighting the multitude of relations between reflection groups, algebraic geometry and singularity theory, we refer to the survey papers by I. Dolgachev [16], A. Givental [25] and O. Shcherbak [55].

In this paper we will be concerned here with the three reflection groups $G = \mathbb{T}, \mathbb{O}, \mathbb{I}$ of the tetrahedron, octahedron and icosahedron respectively and leave the dihedral cases corresponding to $(2,2,n)$ aside.

As abstract groups, the rotations groups $S\mathbb{T}, S\mathbb{O}, S\mathbb{I}$ are isomorphic to the alternating group $\mathcal{A}_4$, the symmetric group $\mathcal{S}_4$ and the alternating group $\mathcal{A}_5$.

1.2. **The Goursat Invariants.** In his memoir [27], Goursat gives an account of the *invariant theory* of the rotation and reflection groups of the regular polyhedra and studies in some detail the beautiful surfaces in three space invariant under these groups. W. Barth's celebrated 65 nodal sextic [6] is a special member of a

Polyhedron	Triangle type	Group	Order	Coxeter type	ADE type
Tetrahedron	$(2,3,3)$	$\mathbb{T}$	24	A_3 $\bullet\!-\!\bullet\!-\!\bullet$	E_6
Octahedron	$(2,3,4)$	$\mathbb{O}$	48	B_3 $\bullet\!-\!\bullet\overset{4}{-}\!\bullet$	E_7
Icosahedron	$(2,3,5)$	$\mathbb{I}$	120	H_3 $\bullet\overset{5}{-}\!\bullet\!-\!\bullet$	E_8

TABLE 1. Basic data for the Platonic reflection groups: spherical triangle triples, group orders, Coxeter types, and corresponding ADE types.

family of sextics described in Goursats paper. This work is complementary to the earlier work of F. Klein [37], which is mainly concerned with the invariants of the associated binary groups and the resulting surface singularities now commonly named after him.

Proposition (Goursat): *The ring of invariants of the reflection groups* $G = \mathbb{T}, \mathbb{O}, \mathbb{I}$ *in* $\mathbb{R}^3$ *is freely generated by three polynomials* P, Q, R*:*

$$\mathbb{Q}[x, y, z]^G = \mathbb{Q}[P, Q, R].$$

In fact Goursat constructs a very specific set of generators P, Q and R. In all cases, the invariant of lowest degree is $P = x^2 + y^2 + z^2$ whereas the invariants Q and R are made up by products of linear forms defining symmetry planes permuted by G. He observes that the intersection of the level sets $P = \alpha$, $Q = \beta$, $R = \gamma$ are the orbits of G and consist of $|G| = deg(P) \cdot deg(Q) \cdot deg(R)$ points, from which he concludes that any symmetric polynomial is uniquely expressed polynomially in terms of P, Q, R. Nowadays that result is seen as a special case of the theorem of Shepard and Todd [54], which states that the same holds for any complex reflection group.

There is also an additional *semi-invariant* S, a product of an *odd* number of symmetry planes permuted by the group G, so that it is multiplied by -1 under a reflection in G, and hence is only an invariant of the associated rotation group SG.

For example, for the icosahedron, the sextic Q is the product of the six planes parallel to the opposite pairs of faces of the dual dodecahedron and the dectic R is the product of the ten planes parallel to the pair of faces of the icosahedron. The semi-invariant S has degree 15 and is obtained as product of the 15 reflection planes determined by the 15 pairs of opposite edges of the icosahedron. The following table summarises these facts.

Polyhedron	**Triangle type**	**Group**	**Order**	**deg P**	**deg Q**	**deg R**	**deg S**
Tetrahedron	$(2,3,3)$	$\mathbb{T}$	24	2	3	4	6
Octahedron	$(2,3,4)$	$\mathbb{O}$	48	2	4	6	9
Icosahedron	$(2,3,5)$	$\mathbb{I}$	120	2	6	10	15

TABLE 2. Spherical triangle data, group orders, and degrees of the associated polynomials P, Q, R, and S for the Platonic reflection groups. redundante eintrÃ¤ge, stehen schon in der oberen tabelle

We refer to Appendix A for the explicit form of the Goursat invariants Q, R, S that were used in this paper.

1.3. **Level curves on the sphere.** The surface determined by the equation $P = r$, $r > 0$ an arbitrary fixed value, is the unit sphere in $\mathbb{R}^3$. The level sets of the second invariant $Q = t$, $t \in \mathbb{R}$ cut out a family of algebraic curves

$$C_{r,t}(\mathbb{R})^\circ := \{(x, y, z) \in \mathbb{R}^3 \mid P = r, Q = t\},$$

which belong to the simplest objects that exhibit the symmetry of the corresponding regular polyhedron (Figure 3).

In the sequel it will be convenient to complexify all spaces involved and consider the curve $C_{r,t} \subset \mathbb{P}^3$, the projective closure of the complex points $C_{r,t}(\mathbb{C})^\circ$ of the above curve. In this setting we allow values of $r \neq 0$.

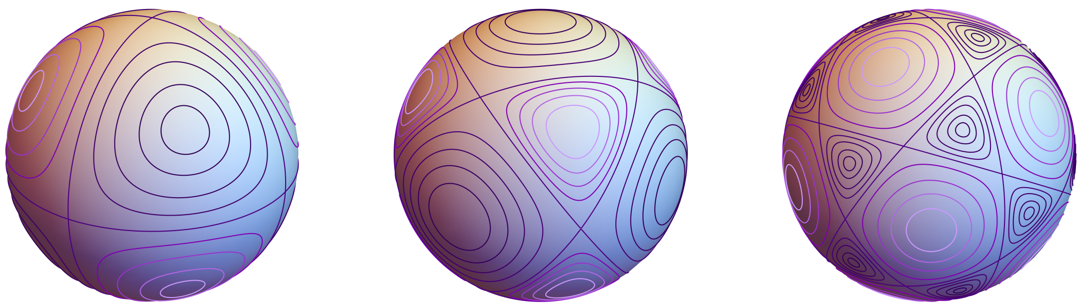
Proposition: *For a general value of t, the curve $C_{r,t}$ is an algebraic curve of genus 4, 9 and 25 for tetrahedron, octahedron and icosahedron respectively.*

There are three values of t for which the curves $C_{r,t}$ are singular, corresponding to the (real) minima, saddles and maxima of the polynomial Q restricted to the sphere $P = r > 0$.

proof: From the adjunction formula it follows that the intersection of a quadric with a surface of degree n is a curve of genus $(n-1)^2$. For $\mathbb{T}, \mathbb{O}, \mathbb{I}$ the degree of Q is $3, 4, 6$ respectively.

Another way of seeing this is by computing the Euler-characteristic of C_0 from the intersection of the sphere with $Q = 0$, which consist respectively of $3, 4, 6$ circles which, projectively and over $\mathbb{C}$, are 2-spheres. As two circles intersect in two antipodal points, we have $6 = 2 \cdot 3, 12 = 3 \cdot 4, 30 = 5 \cdot 6$ intersection points respectively. Hence the reducible curve consist respectively of $3, 4, 6$ copies of $\mathbb{P}^1$, with these intersection points as singular points. For the Euler characteristic of the smoothing $C_{r,t}$ of $C_{r,0}$ we find $2 \cdot 3 - 2 \cdot 6 = -6, 2 \cdot 4 - 2 \cdot 12 = -16, 2 \cdot 6 - 2 \cdot 30 = -48$, respectively, which gives $g = 4, 9, 25$ by equating this to $2 - 2g$. QED

The spherical harmonics associated to the these reflection groups are studied in applied sciences (chemistry and physics in particular) and pure mathematics alike. The papers [41], [5], [32], [38] contain further information.



Curves of genus 4 — Curves of genus 9 — Curves of genus 25

FIGURE 3. Level curves on the sphere $P = r$, $r > 0$, cut out by the invariant Q.

1.4. **Invariants of the rotation groups.** Clearly, the polynomials P, Q, R and S are invariants of the rotation groups SG. But the square S^2 of the semi-invariant S is invariant under the reflection group, so can be expressed polynomially in terms of P, Q, R, i.e. we have a relation of the form

$$S^2 = F(P, Q, R)$$

for some unique polynomial F. A direct consequence is the following:

Proposition: *The ring of invariants of the rotation subgroup $SG := G \cap SO(3)$ is a hypersurface ring of the form*

$$\mathbb{Q}[x, y, z]^{SG} = \mathbb{Q}[P, Q, R, S]/(S^2 - F(P, Q, R))$$

where the relation is given as follows:

Tetrahedron:

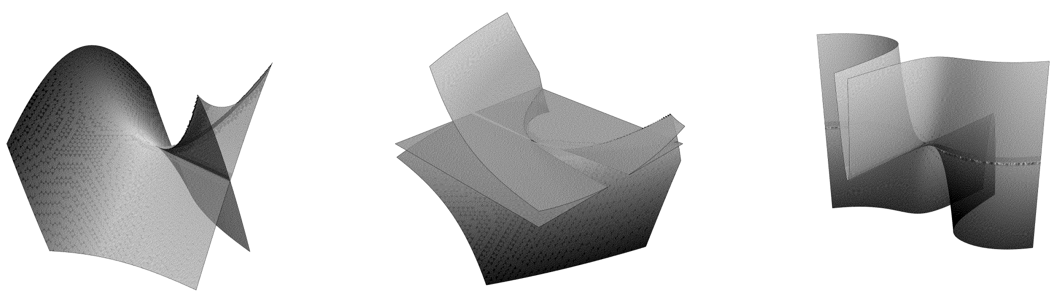
$$S^2 = F(P, Q, R) := -4R^3 + P^2R^2 + 18PQ^2R - (4P^3 + 27Q^2)Q^2 \tag{1}$$

Octahedron:

$$S^2 = F(P,Q,R) := -27R^3 + P(18Q - 4P^2)R^2 + Q^2(P^2 - 4Q)R \tag{2}$$

Icosahedron:

$$\begin{aligned} S^2 = F(P;Q,R) := -R^3 + P^2(4P^3 - 65Q)R^2 + PQ(720Q^2 + 200P^6 - 795P^3Q)R \\ +500P^9Q^2 - 2275P^6Q^3 + 3440P^3Q^4 - 1728Q^5. \end{aligned} \tag{3}$$



Tetrahedral case Octahedral case Icosahedral case

FIGURE 4. Discriminant surfaces of the platonic reflection groups.

Figure 4 shows the corresponding *discriminant surfaces* $F(P,Q,R) = 0$, image of the reflection hyperplanes under the quotient by the reflection groups $G = \mathbb{T}, \mathbb{O}, \mathbb{I}$.

These relations are not recorded in the paper by Goursat, but were determined independently by different authors. We mention [55], but in fact, prior to Klein and Goursat, Brioschi [11] computed the discriminant of the binary sextic, which coincides with the discriminant in the icosahedral case. Certainly, these relations are now routinely obtained by elimination with the help of a computer algebra system, but these are of little help to understand their true nature. The discriminant of the icosahedral reflection group became known as the *caustique mystique* [8], as it was discovered to appear in the "obstacle problem", see [43] for more details.

1.5. **Elliptic threefolds.** It is a remarkable fact that these hypersurfaces can be seen as a *Weierstrass equation in the variables R and S*, thus defining a family of elliptic curves, with P and Q as parameters. In other words, the hypersurface given by the equation $0 = S^2 - F(P, Q, R)$ is naturally an *elliptically fibred threefold*, with the P, Q-plane as basis.

Corollary: *The quotient of the curve $C_{r,t}$ by the rotation group SG is the elliptic curve $E_{r,t}$ in the $S - R$-plane, defined by the equation, specialised to $P = r$ and $Q = t$:*

$$E_{r,t} : S^2 = F(r, t, R), \tag{4}$$

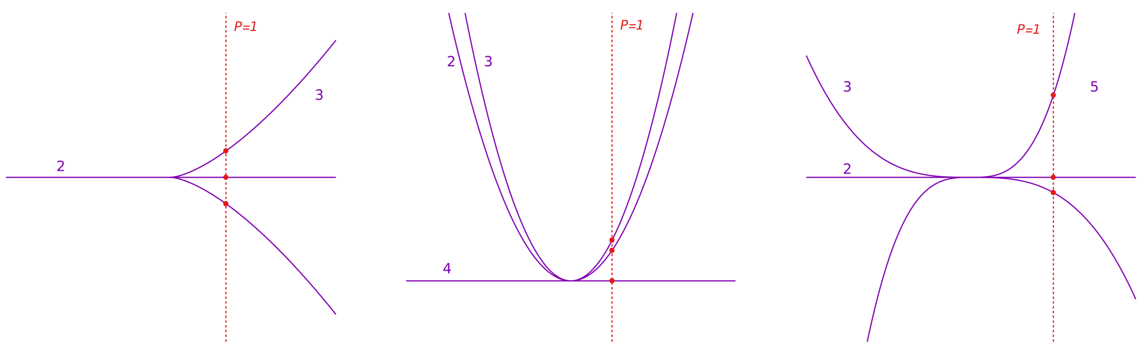
with $F(r, t, R) =$(1),(2) *respectively* (3).

The singular members in this family are determined by the vanishing of the discriminant Δ of $F(P, Q, R)$, seen as cubic in R (Figure 5). These discriminants are easily computed:

Tetrahedron (2,3,3): $\Delta = 2^4 Q^2 (P^3 - 27Q^2)^3$.

Octahedron (2,3,4): $\Delta = 2^4 Q^4 (P^2 - 4Q)^2 (P^2 - 3Q)^3$.

Icosahedron (2,3,5): $\Delta = 2^{12} Q^2 (5P^3 + 27Q)^3 (P^3 - Q)^5$.



Tetrahedral case Octahedral case Icosahedral case

FIGURE 5. Discriminant curves $\Delta = 0$ of $F(P, Q, R)$ in the P-Q plane

1.6. **The platonic elliptic surfaces.** We see that for $P = r$, $r \neq 0$, $Q = t$ the discriminant Δ is of degree $8, 9, 10$ in t, with three roots of multiplicities $2, 3, 3$ for $\mathbb{T}$, $2, 3, 4$ for $\mathbb{O}$ and $2, 3, 5$ for $\mathbb{I}$, thus recovering the platonic triples as multiplicities of the discriminant at the intersections points with P.

Figure 5 shows the above discriminants with dotted line $P = 1$, giving 3 intersection points in the respective cases.

We will see that it is natural to extend these to families of (generalised) elliptic curves over $\mathbb{P}^1$, that is, consider them as *elliptic surfaces*.

The focus on the sections $P = r$, where r is a fixed non-zero value, is quite special. There are some other interesting sections by lines in the $P - Q$ plane that we will comment on in section 7.

2. Elliptic surfaces

By an *elliptic surface* over a curve C we mean a smooth projective surface $\mathcal{E}$ equipped with a proper morphism $\pi : \mathcal{E} \longrightarrow C$, the general fibre of which is a curve of genus 1. We always assume that π is *relatively minimal*, i.e. having no (-1)-curves contracted by the map π. Furthermore, we will assume that a section $\sigma : C \longrightarrow \mathcal{E}$ of π is chosen, so that the general fibre has an origin and becomes an elliptic curve. The set of all sections then aquires the structure of an abelian group and is called the *Mordell-Weil group* $MW(\mathcal{E})$ of the elliptic surface. The intersection of curves on $\mathcal{E}$ make it into the *Mordell-Weil lattice* [48]. Here we will only be dealing with elliptic surfaces over $C = \mathbb{P}^1$.

The elucidation of the basic structure of elliptic surfaces and their degenerate fibres (sometimes called generalised elliptic curves) goes back to Kodaira [39]. For a modern account we refer to the lecture notes of Miranda [45] and the text book of Shioda and Schütt [56].

Elliptic surfaces can be described by *Weierstrass models*

$$y^2 = x^3 + a(t)x + b(t), \text{ or } y^2 = 4x^3 - g_2(t)x - g_3(t),$$

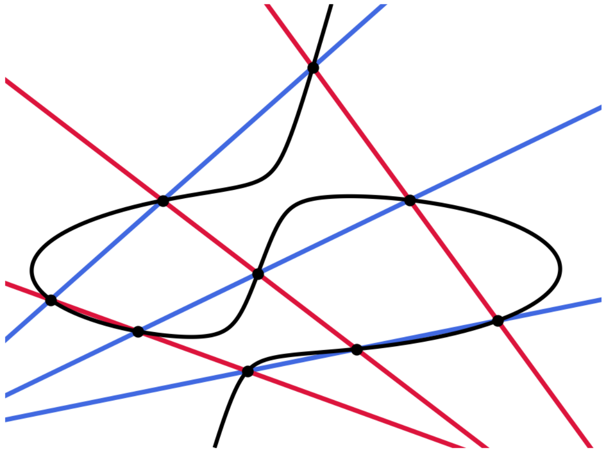
but often this is not the most natural form.

The singular fibres are determined by the vanishing of the discriminant $\Delta := \Delta(t) = g_2(t)^3 - 27g_3(t)^2$; the J-map $J : \mathbb{P}^1 \longrightarrow \mathbb{P}^1$; $t \mapsto g_2^3/\Delta$ assigns to t the absolute invariant $J(t)$ that determines (if $\Delta(t) \neq 0$) the elliptic curves $E_t := \pi^{-1}(t)$ up to $\mathbb{C}$-isomorphism. Note that if we set $S := J^{-1}(0, 1, \infty)$, the restriction $\mathbb{P}^1 \setminus S \longrightarrow \mathbb{P}^1 \setminus \{0, 1, \infty\} = \mathbb{C} \setminus \{0, 1\}$ is unramified. If we write $E_t = \mathbb{C}/\mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H} = \{z \in \mathbb{C} | Im(z) > 0\}$, then τ is determined up to a $PSL_2(\mathbb{Z})$-transformation. More precisely, we have a map

$$J_* : \pi_1(\mathbb{P}^1 \setminus S) \longrightarrow \pi_1(\mathbb{C} \setminus \{0, 1\}) \xrightarrow{\alpha} PSL_2(\mathbb{Z})$$

(see [45], pp.61). We call the image $Im(J_*)$ in the modular group $\Gamma := PSL_2(\mathbb{Z})$ the *(projective) monodromy group*.

Figure 6. Pencil of cubics spanned by two triples of lines.



2.1. **Special elliptic surfaces.** Many studies have been made of special classes of elliptic surfaces. In this paper we will be dealing only with *rational elliptic surfaces* by which one means that $\mathcal{E}$ is a rational surface, i.e. birational to $\mathbb{P}^2$. The geometrically simplest examples arise from *pencils of cubics*

$$tf(x,y,z) - sg(x,y,z) = 0$$

passing through 9 points by blowing up the 9 base points defined by the vanishing of f and g.

For two general cubics, such a surface will have 12 singular fibres of Kodaira type I_1, i.e. ordinary double points. The preimage of the 9 base points are sections of the fibration and one of them can be chosen as origin of the group law on the fibres; the remaining 8 sections are generators for the Mordell-Weil group $MW(\mathcal{E})$. The intersection form on divisors gives the Mordell-Weil group the structure of a *lattice*, which turns out to be of type E_8. By special choice of the cubics f, g one can produce surfaces with fewer singular fibres of various Kodaira types (and other types of Mordell-Weil groups). Among these the fibres of type I_n, which consists of a cycle of n-rational curves are most common and are also called semi-stable. If $e(E_p)$ denotes the Euler number of the fibre $E_t := \pi^{-1}(t)$, $t \in \mathbb{P}^1$, then one has $12 = \sum_{t\in\mathbb{P}^1} e(E_t)$, giving the simplest sort of combinatorial restriction on the possible distribution of singular fibres appearing on a rational elliptic surface.

U. Schmickler-Hirzebruch [53] compiled the list of rational elliptic surfaces with *three singular* fibres and determined the special hypergeometric series that arise as Picard-Fuchs operators attached to these surfaces. Some remarkable surfaces are those related to the so-called *euclidean tuples* $2,2,2,2$; $3,3,3$; $2,4,4$ and $2,3,6$, whose reciprocals add up to 1.

Singular fibres	**Tuple**	**MW group**	**Modular group**	**Dynkin type**
$I_2\, I_2\, I_2^*$	–	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$	$\Gamma(2)$	–
$I_1\, I_1\, I_4^*$	–	$\mathbb{Z}/4\mathbb{Z}$	$\Gamma_1(4)$	–
$I_1\, I_4\, I_1^*$	$(2,2,2,2)$	$\mathbb{Z}/4\mathbb{Z}$	$\Gamma_0(4)$	$\widetilde{D}_5$
$I_1\, I_3\, IV^*$	$(3,3,3)$	$\mathbb{Z}/3\mathbb{Z}$	$\Gamma_0(3)$	$\widetilde{E}_6$
$I_1\, I_2\, III^*$	$(2,4,4)$	$\mathbb{Z}/2\mathbb{Z}$	$\Gamma_0(2)$	$\widetilde{E}_7$
$I_1\, I_1\, II^*$	$(2,3,6)$	0	$\Gamma_0(1)$	$\widetilde{E}_8$

Table 3. Elliptic surfaces attached to the euclidean tuples.

The first surface $I_2 I_2 I_2^*$ is the *Legendre family of elliptic curves* given by the affine equation $y^2 = x(x-1)(x-t)$. The surfaces $I_1\ I_4\ I_1^*$ and $I_1\ I_1\ I_4^*$ are isogenous companions of the Legendre family. The last three surfaces pull-back to a surface with the nice equation $x^p + y^q + z^r - sxyz = 0$ under a cyclic covering $t = s^3, s^4, s^6$. Note that the starred Kodaira fibres appearing here are exactly the extended Dynkin diagrams of type D_5, E_6, E_7 and E_8.

This was followed by the work by Beauville [7], who gave a description of the six cases with four semi-stable singular fibres, now often called *Beauville surfaces*. For convenience of the reader we include a list of the cases:

For these the degree of the J-map is 12, which coincides with the index of the monodromy group in Γ. These surfaces were found independently by C. Schoen [52]. All these surfaces have finite Mordell-Weil group and appeared subsequently in the list of *extremal surfaces* in the work of Persson-Miranda [44]. The number of papers related to these examples is large and quite impossible to overview here.

The classification was pushed further by Herfürtner [31], who compiled the complete list of all surfaces with four singular fibres, given in terms of the Weierstrass covariants $g_2(t)$ and $g_3(t)$. At the same time, U. Persson [49] and R. Miranda [46] gave the complete list of all 279 possible combinations of Kodaira

Singular fibres	**Mordell–Weil group**	**Modular group**
$I_3\, I_3\, I_3\, I_3$	$\mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$	$\Gamma(3)$
$I_4\, I_4\, I_2\, I_2$	$\mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$	$\Gamma_0(4) \cap \Gamma(2)$
$I_5\, I_5\, I_1\, I_1$	$\mathbb{Z}/5\mathbb{Z}$	$\Gamma_1(5)$
$I_6\, I_3\, I_2\, I_1$	$\mathbb{Z}/6\mathbb{Z}$	$\Gamma_0(6)$
$I_8\, I_2\, I_1\, I_1$	$\mathbb{Z}/4\mathbb{Z}$	$\Gamma_0(8)$
$I_9\, I_1\, I_1\, I_1$	$\mathbb{Z}/3\mathbb{Z}$	$\Gamma_0(9)$

TABLE 4. The Beauville elliptic surfaces with four semi-simple fibres

fibres that may occur on a rational elliptic surface. In a precise sense, all cases with up to 12 singular fibres of type I_1, can be obtained by deformation from surfaces with ≤ 4 fibres, see [46].

2.2. **The platonic elliptic surfaces.** What about the three elliptic surfaces obtained in section 1.6 from the Goursat invariants of the 3-dimensional reflection groups?

Proposition: *The fibre types of the three platonic elliptic surfaces* (4) *are*

Polyhedron	**Singular fibres**			
Tetrahedron	$I_2(0)$	$I_3((r/3)^{3/2})$	$I_3(-(r/3)^{3/2})$	$IV(\infty)$
Octahedron	$I_2(r^2/4)$	$I_3(r^2/3)$	$I_4(0)$	$III(\infty)$
Icosahedron	$I_2(0)$	$I_3(-5r^3/27)$	$I_5(r^3)$	$II(\infty)$

TABLE 5. The platonic elliptic surfaces

proof: As we can read off $g_2(t)$ and $g_3(t)$ from the Goursat equations (4), this is a straightforward calculation by hand, PARI or MAGMA. We already have seen that the finite singular fibres arose from a transverse intersection of the discriminant. The multiplicities of the components were exactly the platonic triples (p, q, r) and thus give semi-stable fibres $I_p,\ I_q,\ I_r$. By analysing the fibre over $t = \infty$ we find the result as stated. $\diamond$

So we see we nicely recover the platonic triples in the fibre types; the fibres II (rational curve with cusp), III (two tangent rational curves), IV (three rational curves intersecting in a single point) appearing at infinity are precisely the three other reduced Kodaira-fibres. Their starred Kodaira-duals IV^*, III^*, II^* correspond to Kodaira fibres that consist of curve configurations of the extended E_6, E_7 and E_8 graphs respectively, which are precisely the Dynkin labels corresponding to $\mathbb{T}, \mathbb{O}, \mathbb{I}$!

As rational elliptic surfaces with four singular fibres, these surfaces appear in the list compiled by Herfürtner [31]. Indeed, our three surfaces belong to the group of the 11 rational elliptic surfaces with *reduced fibres only, one of which is unstable*. We call them the "cousins" of our platonic elliptic surfaces and we obtain in this way a rather strange *platonic family* of elliptic surfaces of which it is not clear how the members are related precisely.

Case	**Fibre types**				**MW lattice**	**Nm. in** [48]	$\mathbf{deg}(J)$
Tetrahedron	2	3	3	IV	$\langle 1/6\rangle \oplus \mathbb{Z}/3\mathbb{Z}$	61	8
	1	2	5	IV	$\langle 1/30\rangle$	56	8
	1	1	6	IV	$\langle 1/2\rangle \oplus \mathbb{Z}/3\mathbb{Z}$	51	8
Octahedron	2	3	4	III	$\langle 1/12\rangle \oplus \mathbb{Z}/2\mathbb{Z}$	59	9
	1	3	5	III	$\langle 1/30\rangle$	56	9
	1	2	6	III	$\langle 1/6\rangle \oplus \mathbb{Z}/2\mathbb{Z}$	53	9
	1	1	7	III	$\langle 1/14\rangle$	47	9
Icosahedron	2	3	5	II	$\langle 1/30\rangle$	56	10
	1	4	5	II	$\langle 1/20\rangle$	55	10
	1	2	7	II	$\langle 1/14\rangle$	47	10
	1	1	8	II	$\langle 1/8\rangle$	45	10

TABLE 6. The platonic family

(Here $\langle m\rangle$ means $\mathbb{Z}\sigma$, with $\sigma^2 = m$).

We will refer to these elliptic surfaces by their abbreviated Kodaira-label and speak of "the $127II$ elliptic surface" for the penultimate case in this list. For example for the three platonic cases we refer to the $234III$ elliptic surface as the $\mathbb{O}$-elliptic surface etc.

As in all these 11 cases the Mordell-Weil group is of rank one, we see that there is a non-torsion section. It was remarked by Elkies [23] that the elliptic surface $235II = \mathbb{I}$ is the one with the lowest height $= 1/30$, but its direct connection to the icosahedron was not mentioned there.

2.3. **The Kleinian elliptic surfaces.** There are three further remarkable elliptic surfaces that are naturally attached to the platonic solids via the corresponding *binary* polyhedral groups **T**, **O** and **I**. They can be understood in terms of the Beauville surfaces as follows. The Beauville surface $I_3I_3I_3I_3$, better known as the *Hesse pencil*, has its four singular fibres lying on the vertices of the regular tetrehedron in the 2-sphere, identified with $\mathbb{P}^1$. Three faces of the tetrahedron come together at a vertex, where we find fibres of type I_3. The monodromy group is the full congruence group $\Gamma(3)$ and the surface is identified with the universal elliptic curve with full level 3 structure over the modular curve $X(3) = \mathbb{P}^1$.

When we pull-back the Beauville surface $I_4I_4I_2I_2$ by a degree 2 map $\mathbb{P}^1 \longrightarrow \mathbb{P}^1$ that ramifies at the I_2 fibres, we obtain an elliptic surface with $2 + 2 \times 2 = 6$ fibres of type I_4, which are located on the vertices of a regular octahedron, where 4 faces come together. The monodromy group is the full congruence group $\Gamma(4)$ and the surface can be identified with the universal elliptic curve with full level 4 structure over $X(4) = \mathbb{P}^1$. However, it is no longer a *rational* elliptic surface, as its geometrical genus $p_g = 6 \cdot 4/12 - 1 = 1$; in fact it is a K3 surface.
When we pull-back the Beauville surface $I_5I_5I_1I_1$ by a cyclic degree 5 map $\mathbb{P}^1 \longrightarrow \mathbb{P}^1$ ramified over the I_1 fibres we obtain an elliptic surface with $2 + 5 \times 2 = 12$ fibres of type I_5, which are located on the vertices of a regular icosahedron, where 5 faces come together. The monodromy group is the full congruence group $\Gamma(5)$ and the surface can be identified with the universal elliptic curve with full level 5 structure over the modular curve $X(5) = \mathbb{P}^1$. But of course, this is not a *rational* elliptic surface. In fact $p_g = 12 \cdot 5/12 - 1 = 4$.

These three cases relate to the *binary* groups **G** rather than to the rotation groups SG we were considering earlier. If we denote the extended modular group by $\tilde{\Gamma} := SL_2(\mathbb{Z})$, one has $\mathbf{T} \simeq SL_2(\mathbb{Z}/3) = \tilde{\Gamma}/\tilde{\Gamma}(3)$, $\mathbf{O} \simeq SL_2(\mathbb{Z}/4) = \tilde{\Gamma}/\tilde{\Gamma}(4)$, $\mathbf{I} = SL_2(\mathbb{Z}/5) = \tilde{\Gamma}/\tilde{\Gamma}(5)$. Note that for $n > 5$ the modular curve $X(n)$ is no longer rational, and one obtains elliptic surfaces over curves of higher genus. We refer to [60] for explicit equations for other modular elliptic surfaces.

3. Normalised Period integrals

In fact, it was the systematic study of normal forms of elliptic curves by R. Fricke und F. Klein [37] that led to the study of families of elliptic curves with level structures over modular curves. Fricke and Klein also described in some detail the Picard-Fuchs differential equations, satisfied by the period integrals attached to the surface. We briefly recall the facts, for more information we refer to [58].

3.1. **Pull-back of the universal elliptic curve.**

Writing the equation of an elliptic surface in Weierstrass form as

$$y^2 = 4x^3 - g_2(t)x - g_3(t),$$

we extract as before the discriminant $\Delta(t)$ and the J-function $J(t)$ as

$$\Delta(t) := g_2(t)^3 - 27g_3(t)^2, \quad J(t) := g_2(t)^3/\Delta(t).$$

We can form period integrals by integrating the canonical differential

$$\omega := \frac{dx}{y}, \quad y = \sqrt{4x^3 - g_2(t)x - g_3(t)}$$

over a family of horizontal closed cycles $\gamma_t \in H_1(E_t, \mathbb{Z})$:

$$\phi(t) := \int_{\gamma_t} \omega.$$

However, as explained in [37], sometimes it is useful to use a slightly different normalisation which arises by using the *normalised differential*

$$\Omega := \sqrt{\frac{g_3(t)}{g_2(t)}}\omega,$$

as the corresponding *normalised period integrals*

$$\phi(t) := \int_{\gamma(t)} \Omega = \sqrt{\frac{g_3(t)}{g_2(t)}} \int_{\gamma(t)} \frac{dx}{\sqrt{4x^3 - g_2(t)x - g_3(t)}}$$

now can be expressed in terms of $J = J(t)$ as

$$\int_{\gamma(t)} \frac{dx}{\sqrt{4x^3 - \frac{27J}{J-1}(x+1)}}, \tag{5}$$

and hence can be checked to satisfy the differential equation:

$$\frac{d^2\phi}{dJ^2}+\frac{1}{J}\frac{d\phi}{dJ}+\frac{1}{144}\frac{31J-4}{J^2(J-1)^2}\phi=0. \tag{6}$$

The elliptic curve

$$y^2=4x^3-\frac{27J}{J-1}(x+1)$$

appearing in the integral (5) defines an elliptic surface with singular fibres at 0, 1 and ∞, with fibres of type II, III^* and I_1 respectively. It is called the *universal elliptic curve over the J-line*, as the fibre at J has J as J-invariant and thus each value of J appears precisely once in the family.

The differential equation for the periods of this universal family can be written in the following so-called θ-form:

$$\left(\theta-\frac{1}{12}\right)\left(\theta-\frac{5}{12}\right)-J\theta^2=0,\quad \theta=J\frac{d}{dJ},$$

which we recognise as hypergeometric operator, with extended Riemann symbol

$$\left\{\begin{array}{ccc} II & III^* & I_1 \\ \hline 0 & 1 & \infty \\ \hline 1/12 & 0 & 0 \\ 5/12 & 1/2 & 0 \end{array}\right\}$$

In order to use this to get a differential operator in the original variable t, we have to replace the derivative with respect to J by derivative with respect to t, i.e. we have to set

$$\frac{d}{dJ}=\frac{1}{\partial J/\partial t}\frac{d}{dt}. \tag{7}$$

3.2. Some examples.

Legendre Family. For the Legendre elliptic curve $y^2=x(x-1)(x-t)$ one has

$$g_2(t)=\frac{4}{3}(t^2-t+1),\quad g_3(t)=\frac{4}{27}(2t-1)(t-2)(t+1) \tag{8}$$

and

$$\Delta(t)=16t^2(t-1)^2,\quad J(t)=\frac{4}{27}\frac{(t^2-t+1)^3}{t^2(t-1)^2}. \tag{9}$$

In order to compute the operator in the variable t one has to replace $J(t)$ in (6) by (7), perform a transformation by $\sqrt{g_3(t)/g_2(t)}$ and scale the result. We get the well-known hypergeometric operator

$$\frac{d}{dt}^2 + \frac{2t-1}{t(t-1)}\frac{d}{dt} + \frac{1}{4t(t-1)} \tag{10}$$

or in θ-form

$$\theta^2 - t\left(\theta + \frac{1}{2}\right)^2. \tag{11}$$

The extended Riemann symbol of this operator is

$$\left\{\begin{array}{ccc} I_2 & I_2 & I_2^* \\ \hline 0 & 1 & \infty \\ \hline 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \end{array}\right\}. \tag{12}$$

Apery family. Another example is the Apery-family $y^2 - (t+1)xy - ty = x^3 + tx^2$, better known as the rational elliptic surface with singular fibers of type $I_5I_1I_1I_5$, which has

(13)
$$g_2(t) = \frac{1}{12}(t^4 + 12t^3 + 14t^2 - 12t + 1), \quad g_3(t) = \frac{1}{216}(t^2+1)(t^4 + 18t^3 + 74t^2 - 18t + 1)$$

and

$$\Delta(t) = t^5(-t^2 - t + 1), \quad J(t) = \frac{(t^4 + 12t^3 + 14t^2 - 12t + 1)^3}{1728 \cdot t^5(-t^2 - t + 1)}. \tag{14}$$

As differential equation we get

$$\frac{d}{dt}^2 + \frac{3t^2 + 22t - 1}{t(-t^2 - t + 1)}\frac{d}{dt} - \frac{t+3}{t(-t^2 - t + 1)} \tag{15}$$

or in θ-form

$$\theta^2 - t\left(11\theta^2 + 11\theta + 1\right) - t^2(\theta+1)^2. \tag{16}$$

The extended Riemann symbol of the Apery-operator is

$$\left\{\begin{array}{cccc} I_5 & I_1 & I_1 & I_5 \\ \hline 0 & -\frac{11-5\sqrt{5}}{2} & -\frac{11+5\sqrt{5}}{2} & \infty \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right\}. \tag{17}$$

3.3. **Operators for the platonic surfaces.** If we apply this to the families $\mathcal{E}_G$ ((4)) by putting $P = r$ and $Q = t$ and bringing the cubic in Weierstrass form $4x^3 - g_2(t)x - g_3(t)$, we read off the covariants g_2 and g_3 which then immediately give the differential equations for the normal integrals. The result is:

Proposition: *The Picard-Fuchs differential equations satisfied by the normalised period integral for the elliptic surfaces* $\mathbb{T}, \mathbb{O}, \mathbb{I}$ are

Case	Operator
Tetrahedron	$r^3\theta^2 - 3^3t^2(\theta+2/3)(\theta+4/3)$
Octahedron	$4r^4\theta^2 - r^2t(28\theta^2+28\theta+9) + 3\cdot 4^2t^2(\theta+3/4)(\theta+5/4)$
Icosahedron	$20r^6\theta^2 + r^3t(88\theta^2+88\theta+25) - 3\cdot 6^2t^2(\theta+5/6)(\theta+7/6)$

The extended Riemann symbols of these operators are:

Tetrahedral operator $\mathbb{T}$:

$$\left\{\begin{array}{cccc} I_2 & I_3 & I_3 & IV \\ \hline 0 & (r/3)^{3/2} & -(r/3)^{3/2} & \infty \\ \hline 0 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 4/3 \end{array}\right\}$$

Octahedral operator $\mathbb{O}$:

$$\left\{\begin{array}{cccc} I_4 & I_2 & I_3 & III \\ \hline 0 & r^2/4 & r^2/3 & \infty \\ \hline 0 & 0 & 0 & 3/4 \\ 0 & 0 & 0 & 5/4 \end{array}\right\}$$

Icosahedral operator $\mathbb{I}$:

$$\left\{\begin{array}{cccc} I_2 & I_3 & I_5 & II \\ \hline 0 & -5r^3/27 & r/3 & \infty \\ \hline 0 & 0 & 0 & 5/6 \\ 0 & 0 & 0 & 7/6 \end{array}\right\}$$

The value of r may be put equal to 1 or any other value $\neq 0$: the operators only differ by a scaling of the variable t. We see that the operator for the tetrahedron is

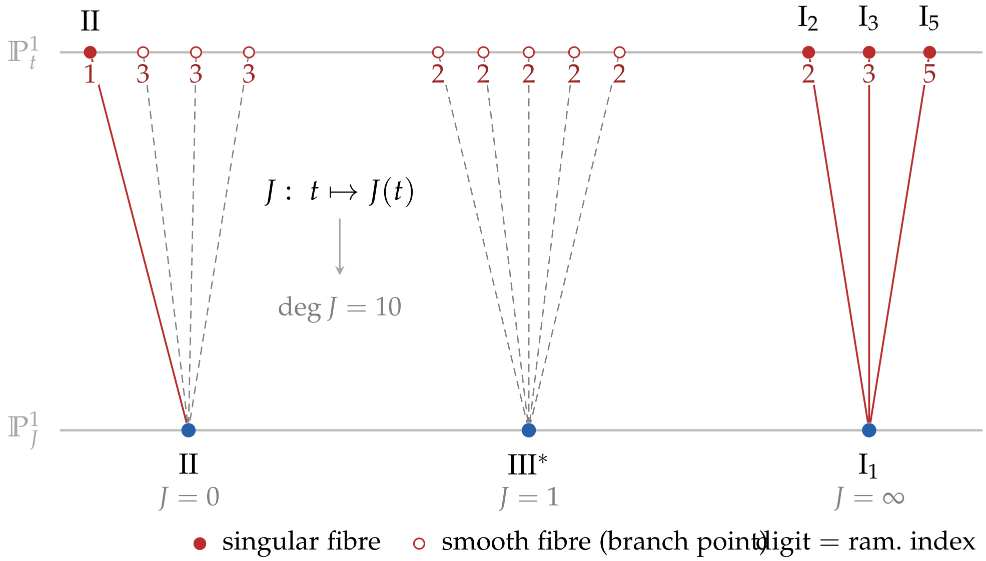


Figure 7. The J-map of the icosahedral surface $I_2\, I_3\, I_5\,$II as a degree-10 branched cover of the J-line. Filled dots are singular fibres of the surface, open dots smooth fibres (branch points of J); the digit at each preimage is its ramification index.

obtained as a quadratic pull-back of a corresponding hypergeometric operator. This operator belongs to the elliptic surface with three singular fibres I_1, I_3, IV^* by a covering map $\mathbb{P}^1 \longrightarrow \mathbb{P}^1$, $t \mapsto t^2$, of degree two ramifying over the I_1 and IV^* fibres.

4. The platonic sequences

4.1. The Apéry case. First discovered in [2], the sequence of (small) Apéry numbers $1, 3, 19, 147, 1251, 11253, \ldots$, whose nth term is the binomial sum

$$A_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}$$

satisfies the two term recursion relation

$$(n+1)^2 A_{n+1} = (11n^2 + 11n + 3)A_n + (n-1)^2 A_{n-1},$$

which implies that the generating series

$$\phi(t) = \sum_{n=0}^{\infty} A_n t^n$$

satisfies the second order differential equation

$$\mathcal{A}\phi(t) = 0, \quad \mathcal{A} := \theta^2 - t(11\theta^2 + 11\theta + 3) - t^2(\theta + 1)^2.$$

It was recognised by F. Beukers [9] that $\mathcal{A}$ is the Picard-Fuchs operator of the Beauville elliptic surface $I_5\ I_1\ I_1\ I_5$, which we already mentioned in Chapter 3, (16), and $\phi(t)$ is the expansion at the unique holomorphic period integral at a I_5 cusp of this surface. This is why the corresponding family of elliptic curves is often called Apery family.

4.2. **Period sequences.** We have seen that the period integrals on any elliptic surface satisfy a second order Picard-Fuchs equation. By a translation we can shift any singularity to the origin, and study the solutions of the differential equation at 0. Near a singularity where the elliptic surface acquires a Kodaira fiber of type I_n, there is a preferred basis of solutions defined on a slit disc around 0, consisting of the *holomorphic solution* ϕ and the *logarithmic solution* ψ. Here $\phi(t)$ expands in a power series, whereas $\psi(t)$ has a logarithmic singularity.

$$\phi(t) = \sum_n a_n t^n, \;\; a_0 = 1, \quad \psi(t) = \phi(t)\log(t) + f(t), \;\; f(t) = \sum_n b_n t^n, \;\; b_0 = 0.$$

A Picard-Fuchs operator of the form ($P_0 \neq 0$, $P_r \neq 0$)

$$P_0(\theta) + tP_1(\theta) + t^2 P_2(\theta) + \cdots + t^r P_r(\theta) \tag{18}$$

is said to be of degree r. The sequence a_n will then satisfy an r-term recursion relation:

$$P_0(n+1)a_{n+1} = P_1(n)a_n + P_2(n-1)a_{n-1} + \cdots + P_r(n-r)a_{n-r}. \tag{19}$$

It is a theorem formulated by Christol [12] that in such a geometrical situation the series $\phi(t)$ has the *Eisenstein property*: there exists a natural number N such that $\phi(Nt) \in \mathbb{Z}[[t]]$, i.e. if we scale the parameter by a factor of N, we obtain a series with integral coefficients.

We call a differential operator and the corresponding holomorphic solution $\phi(t)$ *critically scaled* if $\phi(t) \in \mathbb{Z}[[t]]$, $\phi(t/k) \notin \mathbb{Z}[[t]]$ for all $k > 1$, and if $a_1 \geq 0$.

It is customary for an operator (although not always useful) to replace the parameter t by a parameter $t' = \lambda t$ such that $\phi(t')$ is critically scaled.

In this way we can attach a unique integer sequence $a_0 = 1, a_1, a_2, \ldots$ to each I_n fibre of any elliptic surface. If the Picard-Fuchs operator is of degree 2, the sequence satisfies a 2-term recursion relation.

4.3. **The platonic cases.** We followed this program for each platonic operator, which have three fibres of type I_n. When we shift such a point α to the origin by $t \mapsto t - \alpha$, there is a unique holomorphic solution to the (shifted) and critically scaled Picard-Fuchs equation of the form

$$\phi(t) = \sum_n a_n t^n, \quad a_0 = 1, \quad \phi(t) \in \mathbb{Z}[[t]].$$

Proposition: After critical scaling we obtain the following differential operators and period sequences:

Tetrahedral sequences

$\mathbb{T}(3)$	$1, 12, 198, 3720, 75690, 1626912, 36376704, 837364176, \ldots$	OEIS: no entry, see [59]
(or $23\underline{3}IV$)	$2\theta^2 - 3t(27\theta^2 + 27\theta + 8) + 3^6 t^2 \left(\theta + \frac{2}{3}\right)\left(\theta + \frac{4}{3}\right)$	$I_2\left(\frac{1}{27}\right)$ $I_3\left(\frac{2}{27}\right)$ $I_3(0)$ $IV(\infty)$
$2\underline{3}3IV$	Same as $\mathbb{T}(3)$	Same as $\mathbb{T}(3)$
$\mathbb{T}(2)$	$1, 0, 18, 0, 810, 0, 45360, 0, 2806650, 0, 183891708, 0, \ldots$	OEIS: transformed $A006480$
(or $\underline{2}33IV$)	$\theta^2 - 3^4 t^2 \left(\theta + \frac{2}{3}\right)\left(\theta + \frac{4}{3}\right)$	$I_2(0)$ $I_3\left(\frac{1}{9}\right)$ $I_3\left(-\frac{1}{9}\right)$ $IV(\infty)$

Octahedral sequences

$\mathbb{O}(4)$	$1, 12, 180, 2928, 49860, 875952, 15754704, 288722880, \ldots$	OEIS: $A318245$
(or $23\underline{4}III$)	$3\theta^2 - 4t(28\theta^2 + 28\theta + 9) + 2^{10}t^2\left(\theta + \frac{3}{4}\right)\left(\theta + \frac{5}{4}\right)$	$I_2\left(\frac{3}{64}\right)$ $I_3\left(\frac{1}{16}\right)$ $I_4(0)$ $III(\infty)$
$\mathbb{O}(3)$	$1, 6, 54, 588, 7110, 91476, 1224636, 16849944, 236523078, \ldots$	OEIS: $A186375$
(or $2\underline{3}4III$)	$\theta^2 - 2t(10\theta^2 + 10\theta + 3) + 2^6t^2\left(\theta + \frac{3}{4}\right)\left(\theta + \frac{5}{4}\right)$	$I_2\left(\frac{1}{16}\right)$ $I_3(0)$ $I_4\left(\frac{1}{4}\right)$ $III(\infty)$
$\mathbb{O}(2)$	$1, 12, 612, 25392, 1298340, 67041072, 3633970704, \ldots$	OEIS: no entry
(or $\underline{2}34III$)	$3\theta^2 - 4t(32\theta^2 + 32\theta + 9) - 2^{12}t^2\left(\theta + \frac{3}{4}\right)\left(\theta + \frac{5}{4}\right)$	$I_2(0)$ $I_3\left(\frac{1}{64}\right)$ $I_4\left(-\frac{3}{64}\right)$ $III(\infty)$

Icosahedral sequences

$\mathbb{I}(5)$	$1, 10, 120, 1540, 20500, 279480, 3876600, 54496200, \ldots$	OEIS: $A318495$
(or $23\underline{5}II$)	$2\theta^2 - t(59\theta^2 + 59\theta + 20) + 2^4 3^3 t^2\left(\theta + \frac{5}{6}\right)\left(\theta + \frac{7}{6}\right)$	$I_2\left(\frac{1}{16}\right)$ $I_3\left(\frac{2}{27}\right)$ $I_5(0)$ $II(\infty)$
$\mathbb{I}(3)$	$1, 30, 1440, 85260, 5606100, 391231080, 28360117800, \ldots$	OEIS: $A318496$
(or $2\underline{3}5II$)	$10\theta^2 - 3t(333\theta^2 + 333\theta + 100) + 24 \cdot 3^6 t^2\left(\theta + \frac{5}{6}\right)\left(\theta + \frac{7}{6}\right)$	$I_2\left(\frac{2}{432}\right)$ $I_3(0)$ $I_5\left(\frac{2}{27}\right)$ $II(\infty)$
$\mathbb{I}(2)$	$1, 20, 1140, 68240, 4572100, 321427920, 23420014800, \ldots$	OEIS: no entry
(or $\underline{2}35II$)	$5\theta^2 - 4t(88\theta^2 + 88\theta + 25) - 2^8 3^3 t^2\left(\theta + \frac{5}{6}\right)\left(\theta + \frac{7}{6}\right)$	$I_2(0)$ $I_3\left(\frac{5}{432}\right)$ $I_5\left(-\frac{1}{16}\right)$ $II(\infty)$

TABLE 7. Platonic operators and integral sequences

These operators and most of the integer sequences appear in the work of B.Klee [38]. For the integer sequences associated to the 8 ”cousins” we refer to Appendix B.

4.4. **Laurent polynomial representations.** We say a sequence of integers $a_0, a_1, a_2, \ldots$ has a *Laurent polynomial representation* if there exists a Laurent polynomial $F \in \mathbb{Z}[x, y, 1/x, 1/y]$ such that

$$a_n = [F^n]_0,$$

where $[-]_0$ denotes the operation of taking the constant term. Many interesting sequences of integers admit such a representation, which is rather useful for establishing congruence properties of the numbers appearing in the sequence.

We obtained Laurent polynomial representations for several of the sequences that emerge from the platonic surfaces

Proposition:

The sequence $\mathbb{T}(3)$*:* $1, 12, 198, 3720, 75690, 1626912, 36376704, 837364176, \ldots$ *has Laurent polynomial representation with*

$$F_{\mathbb{T}(3)} = \frac{(y+2+x)(y-2x-1)(x-2y-1)}{xy}. \tag{20}$$

The sequence $\mathbb{T}(2)$*:* $1, 0, 18, 0, 810, 0, 45360, 0, 2806650, 0, 183891708, 0, \ldots$ *has Laurent polynomial representation with*

$$F_{\mathbb{T}(2)} = \frac{3x^2y^2 + x^2 + 3xy^2 + 2x + 1}{xy}. \tag{21}$$

The sequence $\mathbb{O}(4)$*:* $1, 12, 180, 2928, 49860, 875952, 15754704, 288722880, \ldots$ *has Laurent polynomial representation with*

$$F_{\mathbb{O}(4)} = \frac{x^2y^2 + 2x^2y - 3x^2 + 2xy^2 + 12xy + 2x - 3y^2 + 2y + 1}{xy}. \tag{22}$$

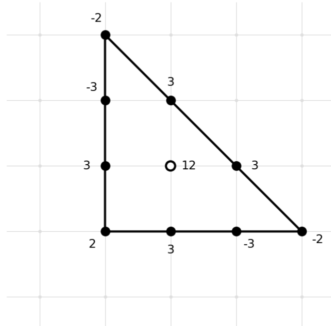
The sequence $\mathbb{O}(3)$*:* $1, 6, 54, 588, 7110, 91476, 1224636, 16849944, 236523078, \ldots$ *has a Laurent polynomial representation with*

$$F_{\mathbb{O}(3)} = \frac{(y+2x+1)(xy+x+2y)}{xy}. \tag{23}$$

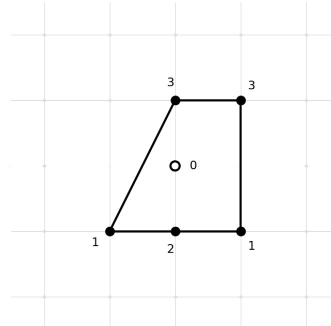
The sequence $\mathbb{I}(5)$*:* $1, 10, 120, 1540, 20500, 279480, 3876600, 54496200, \ldots$ *has a Laurent representation with*

$$F_{\mathbb{I}(5)} = \frac{2x^2y^2 + 3x^2y - 2x^2 - xy^2 + 10xy - 3x + y + 2}{xy}. \tag{24}$$

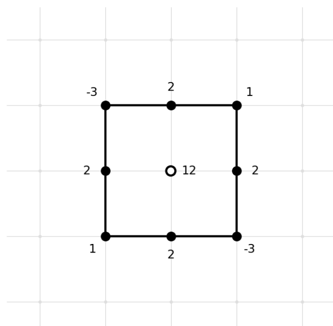
These Laurent polynomial were obtained by starting with one of the *reflexive polygons* [50] and putting general coefficients on the vertices of the polygon to obtain a candidate Laurent polynomial F. Computing the first few powers of F and solving the resulting equations $[F^n]_0 = a_n$ for the first few values of of n



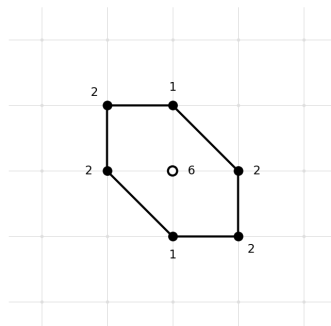
Newton polygon of $F_{\mathbb{T}(3)}$.



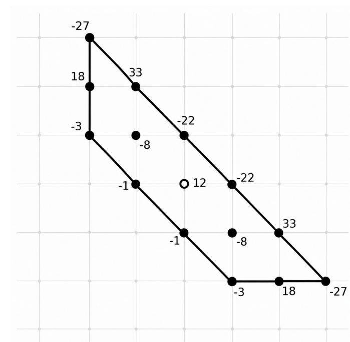
Newton polygon of $F_{\mathbb{T}(2)}$.



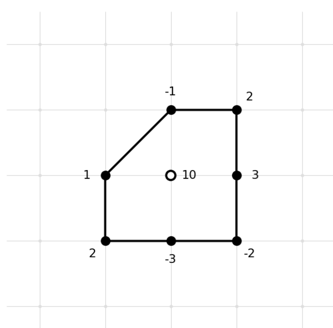
Newton polygon of $F_{\mathbb{O}(4)}$.



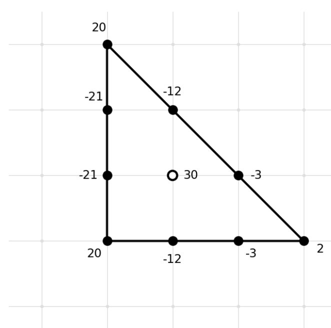
Newton polygon of $F_{\mathbb{O}(3)}$.



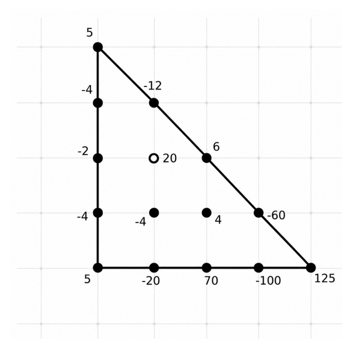
Newton polygon of $F_{\mathbb{O}(2)}$.



Newton polygon of $F_{\mathbb{I}(5)}$.



Newton polygon of $F_{\mathbb{I}(3)}$.



Newton polygon of $F_{\mathbb{I}(2)}$.

FIGURE 8. Newton polygons of the platonic Laurent polynomials

produces candidate Laurent polynomials that then can be verfied to produce the sequence a_n for higher values of n.

A Laurent polynomial F defines a family of curves E_t° in the torus $(\mathbb{C}^*)^2$

$$E_t^\circ := \{(x,y) \in (\mathbb{C}^*)^2 \ \mid \ 1 - tF = 0\}.$$

This curves E_t° can be compactified in $\mathbb{P}^2$ by rationalising and homogenising the equation $1 - tF = 0$. With this geometric viewpoint one can see that the Laurent polynomial generates the integer sequence, by computing the period integral:

$$\phi(t) = \frac{1}{(2\pi i)^2} \oint \frac{1}{1-tF} \frac{\mathrm{d}x\mathrm{d}y}{xy} \tag{25}$$

$$= \sum_{n=0}^{\infty} [F(x,y)^n]_0. \tag{26}$$

4.5. **Elliptic surfaces from Laurent polynomials.** It is also useful to replace t by t/s, so that we end up with a *pencil* over $\mathbb{P}^1 \ni (t:s)$. For example the Laurent polynomial $F_{\mathbb{T}(3)}$ defining the sequence $\mathbb{T}(3)$ leads to a pencil of cubics

$$C(t,s) : t(y + x + 2z)(y - 2x - z)(x - 2y - z) - sxyz = 0.$$

For $(t:s) = (0:1)$ we obtain the coordinate lines $xyz = 0$; whereas for $(t:s) = (1:0)$ we obtain three further lines $(y + x + 2z)(y - 2x - z)(x - 2y - z) = 0$, all passing through the point $(-1:-1:1)$ and intersecting the coordinate triangle in 9 distinct points.

All cubics $C(t,s)$ of the pencil pass through these nine points (Figure 9). There are two further reducible elements in the pencil, namely $(t:s) = (2:27)$: We obtain three further lines

$$(-2y + 2z + x)(2y + z + 2x)(-y - 2z + 2x)$$

and $(t:s) = (1:27)$ where the curve degenerates to a line and a conic:

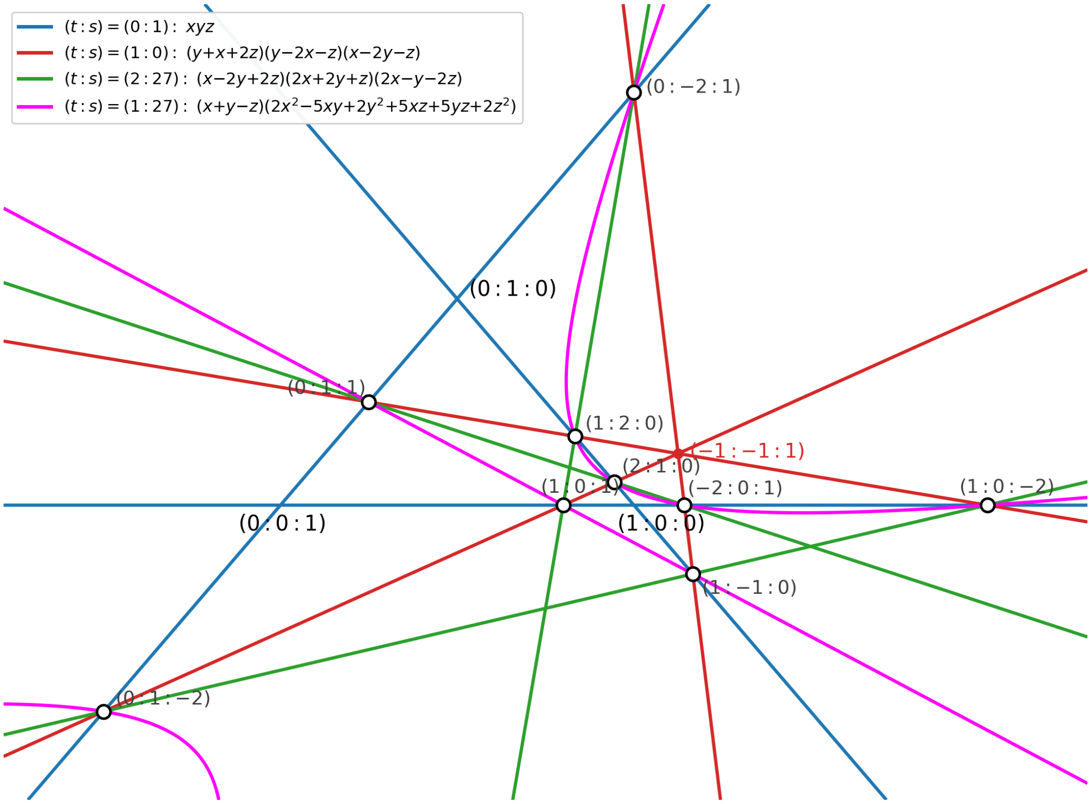
$$(y - z + x)(2x^2 - 5xy + 2y^2 + 5xz + 5yz + 2z^2)$$

$(t:s) = (0:1)$ Three lines (blue) $xyz = 0$ $(t:s) = (1:0)$ Three lines (red) $(y + x + 2z)(y - 2x - 2z)(x - 2y - 2z) = 0$ $(t:s) = (2:27)$ Three lines (green) $(t:s) = (1:27)$ Line and Conic (magenta)

Hence we obtain an elliptic surface with Kodaira fibres I_3 at 0, IV at ∞, I_3 at $2/27$, I_2 at $1/27$. Note that we have made a complete circle: starting from the platonic elliptic surface $233IV$, we determined its Picard-Fuchs equation from which we obtained an integer sequence for which we found a Laurent polynomial, which gave us back the elliptic surface we started with:

$$332IV \rightsquigarrow PF \rightsquigarrow I.S.T1 \rightsquigarrow LPF_1 \rightsquigarrow 332IV$$

Figure 9. Reducible fibers of the pencil
$t \cdot (x + y + 2z)(-2x + y - z)(x - 2y - z) - s \cdot xyz$



Somewhat to our surprise, *this does not happen always*. For the sequence $\mathbb{T}_2$ is turns out that the Laurent polynomial given above defines an elliptic surface of fibre type I_6 at 0, I_1 at $\pm\frac{1}{9}$ and IV at ∞, so here we have

$$233IV \rightsquigarrow PF \rightsquigarrow I.S.T2 \rightsquigarrow LPF_2 \rightsquigarrow 611IV,$$

so that we arrive at a *different* elliptic surface! The issue here is that the surfaces $233IV$ and $613IV$ are connected by degree 3-isogeny, which turns I_2 into I_6 and I_3 into I_1. Two such isogenous surfaces have the same Picard-Fuchs operator and thus produce the same integer sequences.

A similar phenomenon happens for the sequence $O2$ and surfaces $342III$ and $631III$.

$$342III \rightsquigarrow PF \rightsquigarrow I.S.O_2 \rightsquigarrow LPF_4 \rightsquigarrow 621III$$

The two elliptic surfaces are now connected by an isogeny of degree 2.

These isogenies can be read off from Table???, where we can see that the respective surface have torsion sections of order 3 and 2.

4.6. **From quartics to cubics.** The Laurent polynomials F_2, F_3 and F_5 produce pencils of *quartics*. For example, for F_2 we obtain the pencil of quartic curves

$$Q(t,s) : sxyz^2 - t(x+z)(3xy^2z + xz^2 + z^3) = 0$$

The general homogeneous quartic defines a curve of genus 3, but it is easily verified that for general $(t:s) \in \mathbb{P}^1$ the curve $Q(t,s)$ has two ordinary double points located at coordinate points $(0:1:0)$ and $(0:0:1)$, so that the genus of $C(t,s)$ equals 1. These curves can be transformed into cubics by a Cremona transformation. For this one considers the linear system of conics through the two singular points and one of the remaining bases points of the pencil. In concrete terms, after the substitution

$$x = YZ, \quad y = X(X+Z), \quad z = XY$$

in the equation, a factor $X^2Y^2(Z+X)$ can be removed and one obtains a pencil of cubics

$$C(t,s) : sXYZ - t(X+Z)(3XZ + Y^2 + 3Z^2) = 0$$

It is readily verified this produces the alternative Laurent polynomial

$$F_2' := \frac{(X+Z)(3XZ+Y^2+3Z^2)}{XY}$$

for the sequence $\mathbb{T}_2$.

The same can be done for the Laurent polynomial F_3 and F_5. For F_3, we make the substitution

$$x = YZ, \quad y = X(3Z+X), \quad z = XY,$$

which transforms the quartic pencil of F_3 into the cubic pencil

$$D(t,s) := sXYZ - t(3X^2 + XY + 10XZ - YZ + 3Z^2)(Y+Z-X)$$

and hence we get an alternative representation of sequence O_1 by the Laurent polynomial

$$F_3' := \frac{(3X^2 + XY + 10X - Y + 3)(Y + 1 - X)}{XY}$$

In a similar vein, we use the substition

$$x = YZ,\ \ y = X(2Z - X), \quad z = XY$$

to transform the quartic pencil of F_5 into the cubic pencil

$$sXYZ - t(X^2Y + X^2Z - 2XY^2 + 10XYZ - 4XZ^2 - Y^2Z + 3YZ^2 + 4Z^3)$$

and we obtain the corresponding alternative Laurent representation for the sequence $\mathbb{I}_1$.

$$F_5' := \frac{X^2Y + X^2 - 2XY^2 + 10XY - 4X - Y^2 + 3Y + 4}{XY}$$

4.7. **The remaining cases.** What about the remaining sequences $\mathbb{O}(2)$, $\mathbb{I}(3)$ and $\mathbb{I}(2)$. As we have by now aquired a deeper insight into the nature of Laurent representations, we can directly settle case $\mathbb{I}(3)$.

Proposition: *The integer sequence $\mathbb{I}(3)$ does admit a Laurent polynomial representation:*

$$F_{\mathbb{I}(3)} := \frac{2U^3 - 3U^2V - 12UV^2 + 20V^3 - 3U^2 + 30UV - 21V^2 - 12U - 21V + 20}{UV}.$$

proof: We start with the cubic Laurent polynomial F_5' for the sequence $\mathbb{I}_1$. The associated pencil of cubics has six base-points, three of them with multiplicity 2: $(0:1:0), (1:0:0), (2:0:1)$ and three of them with multiplicity 1: $(0:-1:1), (0:4:1), (2:1:0)$. The sequence $\mathbb{I}_2$ is associated to the expansion of the normalised period at $t = 2/27$. The pencil determined by F_5' at $(t:s) = (2:27)$ factorises as

$$(2XY + 2XZ + YZ - 4Z^2)(X - 2Y - 2Z)$$

which defines a line $L : X - 2Y - 2Z = 0$ and a conic $Q : 2XY + 2XZ + YZ - 4Z^2 = 0$, which intersect at ?? We now apply a quadratic Cremona transformation on the three base points $(0:1:0)$, $\ (2:0:1)$, $\ (0:4:1)$. This transforms cubics to cubics ($3 = 2 \times 3 - 3$), the conic Q to a line ($1 = 2 \times 2 - 3$ and the line L into another line ($1 = 2 \times 1 - 1$). The point $(2:0:1)$ is blown up and produces a third line, so after the transformation we have three lines over $2/27$. To compute it, we first replace X be $X + 2Z$ and then make the substitution $X = VW, \quad Y = WU, \quad Z = UV$. After dividing by UVW we obtain a cubic pencil which for $t = 2, s = 27$ factors into three linear factors. Now we make the substitution $t \mapsto 27t + 2s, s \mapsto 27s$, shifting the $2/27$ fibre to 0. By a linear transformation in U, V, W we transform the pencil into the form

$$\begin{aligned} t(2U^3 - 3U^2V - 12UV^2 + 20V^3 - 3U^2W + 30UVW - 21V^2W \\ -12UW^2 - 21VW^2 + 20W^3) - sUVW = 0 \end{aligned}$$

So we obtain the desired Laurent polynomial. $\diamond$.

For the remaining two cases $\mathbb{O}(2)$ and $\mathbb{I}(3)$ there is no Laurent polynomial representation such that the genus of F is equal to 1. With the help of ChatGPT and Claude we were able to find higher genus Laurent polynomials.

Proposition:

The sequence $\mathbb{O}(2)$*:* $1, 12, 612, 25392, 1298340, 67041072, 3633970704, \ldots$ *has Laurent polynomial representation with*

$$F_{\mathbb{O}(2)} = -\frac{\left(3x^2 - 2xy - x + 3y^2 - y\right)\left(9x^3 - 5x^2y - 3x^2 - 5xy^2 + 2xy + 9y^3 - 3y^2\right)}{x^2y^2}. \tag{27}$$

The sequence $\mathbb{I}(2)$*:* $1, 20, 1140, 68240, 4572100, 321427920, 23420014800, \ldots$ *has Laurent polynomial representation with*

$$F_{\mathbb{I}(2)} = \frac{\begin{array}{c} 125x^4 - 60x^3y - 100x^3 + 6x^2y^2 + 4x^2y + 70x^2 - 12xy^3 \\ \hline +20xy^2 - 4xy - 20x + 5y^4 - 4y^3 - 2y^2 - 4y + 5 \end{array}}{xy^2}. \tag{28}$$

5. Some arithmetical aspects

We saw that the integer sequences stemming from the expansion of holomorphic periods of our elliptic surfaces have strong arithmetical properties like the Lucas congruences. In this chapter we will develop this topic further and we will see how one can use the p-adic properties of series expansions of the holomorphic and logarithmic solution to determine a *Frobenius structure*.

An elliptic surface over $\mathbb{P}^1$ determines an elliptic curve over the function field $\mathbb{C}(t)$. Conversely, any elliptic curve over $\mathbb{C}(t)$ has an elliptic surface as minimal model, so that these two notions are basically the same. In all our examples it appeared that the equations we are dealing with have coefficients in $\mathbb{Q}$, so that we have elliptic surfaces over $\mathbb{Q}$ and corresponding elliptic curves over $\mathbb{Q}(t)$. By 'clearing out denominators' we in fact can arrange having an elliptic curve defined over the 2-dimensional ring $\mathbb{Z}[t]$. In algebraic geometric terms we have a family

$$\mathcal{E} \longrightarrow Spec(\mathbb{Z}[t])$$

whose general fibre is an elliptic curve, so $\mathcal{E}$ is an *arithmetic threefold*. There are two sorts of specialisations: We can fix a value of t and obtain, apart from finitely many singular values, an elliptic curve over $Spec(\mathbb{Z})$. Or we can fix a prime p, reduce modulo p and get an elliptic surface over $\mathbb{F}_p[t]$. When we fix both t and p, we get an elliptic curve $E_{t,p}$ over the finite field $\mathbb{F}_p$. The arithmetical properties of elliptic curves is a huge, beautiful and deep subject. We try to retrace the historical lines of discovery and will use only the most basic notions and refer to the textbooks, e.g. [57], for more information.

5.1. **Hasse, Witt and Deuring.** It was known after the works of E. Artin and F.K. Schmidt that the Zeta-function of a curve C of genus g over a finite field $\mathbb{F}_q$, $q = p^f$ factored in a specific way:

$$Z(C;T) := \exp(\sum_n N_k \frac{T^n}{k}) = \frac{P(T)}{(1-T)(1-qT)},$$

was conjectured that an analogue of the Riemann hypothesis holds true: the reciprocal roots of the polynomial P all have size $\sqrt{q}$. For an elliptic curve ($g = 1$) one has

$$P(T) = 1 - a_p T + qT^2$$

where $\#C(\mathbb{F}_q) = 1 - a_p + p$ and the conjecture is equivalent to $|a_p| \leq 2\sqrt{q}$. In series of papers [28],**[?]**,**[?]**, **[?]**, Hasse developed the basic geometrical properties of an elliptic curve E over a (perfect) fields k of characteristic p, purely in the algebraic language of the function field $K = k(E)$. He showed that the group

$$E[n] := \{x \in E(\overline{k}) \mid n \cdot x = 0\}$$

of n-torsion points of E (over the algebraic closure $\overline{k}$) has n^2 elements if n is prime to p, n elements if $n = p^\nu$ and a certain quantity A (now called the Hasse invariant) is non-zero, and 1 element if $n = p^\nu$ and $A = 0$ (now called the super-singular case). Furthermore, he showed that the endomorphism ring $End(E)$ contains $\mathbb{Z}$ as a subring and hence is a ring of characteristic 0. For each (algebraic) endomorphism μ outside the subring $\mathbb{Z}$, he showed that it is *imaginary quadratic*, i.e. satisfies an equation of the form

$$\mu^2 + \nu\mu + m = 0,$$

where $\nu, m \in \mathbb{Z}$ and $\nu^2 \leq m$. He applied this to the Frobenius endomorphism $\pi \in End(E)$, induced by $f \mapsto f^q$ on the function field K and concludes that the Riemann hypothesis holds for elliptic curves over a finite field, i.e. $a_p^2 \leq 4q$. We will call the number a_p *the Frobenuis trace*, as it is interpreted as the trace of the map induced by Frobenius on any Weil cohomology theory.

Over an algebraically closed field k the structure of the torsion points $E[n]$ of E is remarkable: as long as n is prime to p the result is the same as in characteristic zero, forming the well-known pattern of division points in the fundamental period parallelogram of E. But if we specialise to characteristic p, the p^2-torsion points come together in groups of p (if $A \neq 0$) to form the group of p-torsion points $E[p]$ of p elements, and if $A = 0$ all p^2 torsion points specialise to the neutral element, so $E[p]$ consist of one element.

If we divide out a point of order p we obtain a separable, cyclic unramified field extension of an elliptic function field and it is in this context that Hasse [28] introduces his invariant.

In [30] Hasse and Witt reformulate the definition of A and generalise it to curves C of arbitrary genus g over $\mathbb{F}_p$. They start by choosing g k-rational points on C, such that the divisor $D = P_1 + P_2 + \ldots + P_g$ is non-special. For each point they pick a local uniformiser t_i at P_i. They consider the space $P(*D) \subset P(D)$ of principal parts supported at D and the subgroup of those elements of pole order ≤ 1. Furthermore, one considers $K(*D) \subset K(D)$ of elements of the function field with poles in D and the subspace of those elements with pole of order ≤ 1. The factor space $V := P(*D)/K(*D)P(D)/K(D)$ is of dimension g, with the elements r_i, defined by $(r_i)_Q = [1/t_i]$ if $Q = P_i$ and 0 else, as basis. Appling the Frobenius map $f \mapsto f^p$ to r_i and reexpressing the result in terms of the basis r_i leads to a matrix $A = (A_{ij})$:

$$r_i^p = \sum_{j=1}^{g} A_{ij} r_j \in V$$

The A is a $g \times g$ matrix with entries in k, nowadays called *Hasse-Witt matrix*. It is an object of semi-linear algebra, as under a bases change matrix S, the matrix A transforms to $S^{-1}AS^{(p)}$, where $S^{(p)}$ denotes the matrix obtained from S by taking the p-power of each entry. For $g = 1$ the matrix A is a single number and scaling the basis by a factor r changes A to $r^{-1}Ar^p = r^{p-1}A$. If k is algebraically closed we can scale A to 0 or 1; for the prime field $k = \mathbb{F}_p$ A is well defined as element of k. The procedure of Hasse and Witt can be interpreted in newspeak by remarking that the vector space $V = H^1(C, \mathcal{O}_C)$ and A is identified as matrix of the natural

p-linear map $H^1(C, \mathcal{O}_C) \longrightarrow H^1(C, \mathcal{O}_C)$ induced by π, with respect to a basis defined by the divisor D.
[1]

Deuring classified the possible endomorphism rings of elliptic curves in [15] and showed that $A = 0$ is equivalent to $End(E)$ being an order in a quaternion algebra. He computed A for an elliptic curve explicitly with respect to the point at infinity and the uniformiser x/y, using the definition in [30]. If the elliptic curve is given in the form

$$y^2 = f(x) = x^3 + a_2x^2 + a_4x + a_6, \tag{29}$$

the invariant A can be computed as to the coefficient of x^{p-1} in $f^{\frac{p-1}{2}}$. If E is given in Legendre normal form $y^2 = x(x-1)(x-t)$, the Hasse invariant A depends polynomially on the parameter t and Dearing finds (for $p \geq 3$):

$$A = A(t) = (-1)^{\frac{p-1}{2}} \sum_{n=0}^{\frac{p-1}{2}} \binom{\frac{p-1}{2}}{n}^2 t^n, \tag{30}$$

5.2. **Igusa, Manin, Dwork.** We have seen that the Picard-Fuchs operator of the Legendre family $y^2 = x(x-1)(x-t)$ is

$$\mathcal{L} := t(1-t)\frac{d^2}{dt^2} + (1-2t)\frac{d}{dt} - \frac{1}{4}, \tag{31}$$

The unique holomorphic solution to $\mathcal{L}\phi = 0$ near 0 is given by the power series

$$\phi(t) = \sum_{n=0}^{\infty} \binom{\frac{-1}{2}}{n}^2 t^n \tag{32}$$

that is,

$$\phi(t) = 1 + (\frac{1}{2})^2 + (\frac{1\cdot 3}{2\cdot 4})^2 t^2 + (\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6})^2 t^3 \ldots = \sum_{n=0}^{\infty} \binom{2n}{n}^2 (\frac{t}{16})^n$$

Igusa notices in [34] that the expression (30) coincides with the above series, taken modulo p.

$$\phi(t) = A(t) \mod p,$$

[1]The exact sequence of sheaves on C ($\mathcal{O} = \mathcal{O}_C$): $\mathcal{O} \hookrightarrow \mathcal{O}(*D) \twoheadrightarrow \mathcal{O}(*D)\mathcal{O}$ gives by taking global sections $H^1(\mathcal{O}) = Coker(H^0(\mathcal{O}(*D)) \longrightarrow H^0(\mathcal{O}(*D)/\mathcal{O}) = P(*D)/K(*D)) = V$. Similarly one sees from the exact sequence $\mathcal{O} \hookrightarrow \mathcal{O}(D) \twoheadrightarrow \mathcal{O}(D)/\mathcal{O}$ and Riemann-Roch that if D is non-special one has $P(D)/K(D) = H^1(\mathcal{O}) = V$.

so that

$$\mathcal{L}A(t) = 0 \mod p$$

and that as a result of this fact, the roots of the Hasse polynomial $A(t)$ are all distinct.

Manin showed in [51] that in the case $k = \mathbb{F}_p$, i.e. when the invariant A is independent of any choice, we have

$$P(T) \equiv 1 - AT \bmod p \tag{33}$$

where $P(T) = 1 - a_p T + pT^2$ is as before equal to the numerator polynomial of the Zeta-function of the curve. It is strange, but it seems to us that Hasse has overlooked this fundamental fact. Keeping in mind that A is the trace of Frobenius on $H^1(E, \mathcal{O}_E)$, this statement is now an easy consequence of the general Fulton trace formula [24], which was not available at the time.

Together with (30) we get the following:

Theorem For an elliptic curve $E/\mathbb{F}_p$ we have $A \equiv a_p \mod p$. In particular E is supersingular, i.e. $A = 0$, if and only if $a_p \equiv 0 \mod p$. For the Legendre family

$$E_t : y^2 = x(x-1)(x-t) \tag{34}$$

we get at each fiber t, if the reduction E_t mod p is smooth, the trace formula

$$a_p(t) \equiv A(t) \mod p, \tag{35}$$

where $a_p(t)$ is the Frobenius trace of E_t.

Manin shows in [51] that this is not a coincidence for the Legendre family but a more general phenomenon:

Gauss-Manin connection on $H^r_{dR}(E)$

Let A be a k_0-algebra, M an $A-$module with the structure of a $Der_{k_0}(A)-$module. A *linear differential relation* is an equation of the form

$$u_1 m_1 + \cdots + u_g m_g = 0, \tag{36}$$

$u_i \in U(Der_{k_0}(A))$ the universal enveloping algebra, $m_i \in M$.

Let E be an elliptic curve over a field k of characteristic p over the field of constants k_0 with $\dim_k(Der_{k_0}(k)) = 1$, $H^r_{dR}(E)$ the de Rham cohomology, i.e. the cohomology of the complex $\mathcal{O}_E \to \Omega^1_{E/k} \to \ldots$. Each $\delta \in Der_{k_0}(k)$ extends to a connection $\nabla_\delta : H^r_{dR}(E) \to H^r_{dR}(E)$.

The spaces $H^1(E, \mathcal{O}_E)$ and $H^0(E, \Omega^1_E)$ are dual to each other via Serre duality. If r is a generator of $H^1(E, \mathcal{O}_E)$ and $\omega \in H^0(E, \Omega^1_E)$ its dual, then for the Frobenius $F : H^1(E, \mathcal{O}_E) \to H^1(E, \mathcal{O}_E)$ we have

$$A = \langle \omega, F(r) \rangle, \tag{37}$$

where $A \in k$ is the scalar Hasse-Witt matrix of E. We write $\langle \eta, r \rangle$ for the residue form of the Serre-duality pairing. If η is regular, this is the ordinary Serre-duality pairing. In Manin's proof this notation is also used for rational differentials η.

Let $\omega \in H^0(E, \Omega^1_E)$ and $\overline{\omega}$ its image in $H^1_{dR}(E)$.

Theorem (Manin) If $\mathcal{L}(\overline{\omega}) = 0$ is a linear differential relation, then $\mathcal{L}(A) = 0$.

proof sketch: If $\mathcal{L}(\overline{\omega}) = 0$, $\mathcal{L} \in U$, then we have

$$\nabla_{\mathcal{L}} \omega = d\phi, \tag{38}$$

for some function $\phi \in K(E)$. We get

$$\langle \nabla_{\mathcal{L}}(\omega), F(r) \rangle = \langle d\phi, F(r) \rangle, \tag{39}$$

Furthermore we have $C(d\phi) = 0$ for the Cartier operator C and by Cartier's lemma: $\langle \omega, F(r) \rangle = \langle C(\omega), r \rangle^p$. It follows

$$\langle C(\nabla_{\mathcal{L}}(\omega)), r \rangle^p = 0. \tag{40}$$

Manin's Cartier compatibility shows that applying $\mathcal{L}$ to the Hasse–Witt entry is the same as pairing the Cartier transform of $\nabla_{\mathcal{L}}(\omega)$ with r:

$$0 = \langle C(\nabla_{\mathcal{L}}(\omega)), r \rangle^p = \mathcal{L}(\langle C(\omega), r \rangle^p) = \mathcal{L}(\langle \omega, F(r) \rangle) = \mathcal{L}(A). \tag{41}$$

Therefore the Hasse invariant A satisfies the same differential equation as the period class. □

In concrete terms, given a family of elliptic curves $\mathcal{E} \to \mathbb{P}^1$ over $\mathbb{F}_p$, the Picard-Fuchs equation can be used to compute the Frobenius trace at each smooth fiber in the following way:

Lemma If a differential equation ... has a solution in $\overline{\mathbb{F}}_p[[t]]$, then there exists a unique polynomial solution $u \in \overline{\mathbb{F}}_p[t]$ such that $u(0) = 1$ and $\deg(u) < p$. The full set of solutions is given by $\{Q(t^p)u(t) | Q(t) \in \overline{F}_p[[t]]\}$. XXXReference to Honda?

Corollary For a family of elliptic curves $\mathcal{E} \to \mathbb{P}^1$ over $\mathbb{F}_p$, assuming the Picard-Fuchs equation satisfies the conditions of the lemma, we get for the Hasse-invariant

$$A \equiv \left(\sum_{n=0}^{p-1} u_n t^n \right) \cdot Q(t^p) \bmod p, \tag{42}$$

for some $Q(t) \in \overline{\mathbb{F}}_p[[t]]$.

We recall that an elliptic surface $\pi : \mathcal{E} \to \mathbb{P}^1$ fundamental line bundle $\mathcal{L}$, ([45], section II.4) It can be defined as the Hodge bundle $\pi_*(\omega_{\mathcal{E}/\mathbb{P}^1})$, with fibres $H^0(\omega_{E_t})$, and which is dual to $R^1\pi_*(\mathcal{O}_{\mathcal{E}})$, with fibres $H^1(\mathcal{O}_{E_t})$. The degree of this bundle relates to the geometric genus of the elliptic surface ()[45] section IV.1:

$$\deg(\mathcal{L}) = p_g(\mathcal{E}) + 1.$$

For rational elliptic surfaces we have $p_g = 0$, so $\deg(\mathcal{L}) = 1$. Frobenius induces a p-linear map from $\mathcal{L}$ to itself, and as a result the Hasse-invariant can be seen as a section of $\mathcal{L}^{p-1}$, a line bundle of degree $p-1$ Hence the degree of the Hasse polynomial is $\leq p-1$.

Corollary For a family of elliptic curves $\mathcal{E} \to \mathbb{P}^1$ over $\mathbb{F}_p$, assuming the Picard-Fuchs equation satisfies the conditions of the lemma, we get at each smooth fibre t, if the reduction E_t mod p is smooth, the trace formula

$$a_p(t) \equiv \pm \sum_{n=0}^{p-1} u_n t^n \bmod p. \tag{43}$$

We remark that the p dependent sign $\pm$ can not be determined from the Picard-Fuchs operator alone. If we make perform a quadratic twist on the fibres $E_t/\mathbb{F}_p$, then a_p changes by a sign of the quadratic character, by the Picard-Fuchs operator does not change.

5.3. **Tate, Dwork and Katz.** We have seen that for an elliptic curve the Frobenius trace $a_p \mod p$ can be computed by the Hasse invariant and that in a familie is satifies the Picard-Fuchs equation $\bmod p$. It is natural to try to obtain a_p modulo higher powers of p. Of course, as $|a_p| \leq 2\sqrt{p}$, for $p >?$ the residue class mod p uniquely determines the integer a_p so from a practical standpoint this is not very urgent. But for curves of higher genus or higher dimensional varieties and from a theoretical standpoint it turns out to very useful and leads directly to the development of p-adic analysis and p-adic cohomology theories.

The numerator polynomial $P(T) = 1 - a_pT + pT^2 \in \mathbb{Z}$ in general does not factor over $\mathbb{Z}$. If we have a factorisation over the p-adic numbers $\mathbb{Z}_p$,

$$P(T) = (1 - uT)(1 - vT) \in \mathbb{Z}_p[T],$$

then the p-adic order of say u must be 0, that of v be one. One calls $u \in \mathbb{Z}_p$ the unit root. Note that there exists a unit root precisely if the Hasse invariant

$A \neq 0$. XXX Explain? Let $E_t:\ y^2 = x(x-1)(x-t)$ be the Legendre family over $\mathbb{Z}_p[t][(t(1-t))^{-1}]$.

Theorem :Dworks Unit root formula, [19] Let $t_0 \in \mathbb{F}_p$ such that E_{t_0} is smooth and not supersingular. Then the unit root is given by

$$u_{t_0} = (-1)^{(p-1)/2} \frac{\phi(z)}{\phi(z^p)} \,|_{z=t}, \tag{44}$$

where $t \in \mathbb{Z}_p$ the multiplicative lift of t_0.

In [19] Dwork shows that the Zeta-functions of general hypersurfaces varying in a $1-$parameter family over $\mathbb{Q}$ can be restated in terms of endomorphisms of the solution space of the Picard-Fuchs equation. After choosing an embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ is chosen, the family can be considered as family over $\mathbb{Z}_p[t][\Delta(t)^{-1}]$, where $\Delta(t)$ is the discriminant locus. The Picard-Fuchs equation is considered as a $p-$adic differential equation and the formal power series solutions are what Dwork calls $p-$adic cycles, which he develops the theory on.

Frobenius structure on a differential equation

Dwork proves the following lemma, which gives

5.4. **The simplest examples.** First we consider the four well-known hypergeometric families with MUM-point at 0, among which are the Legendre family and the Hesse pencil, and the six Beauville families of elliptic curves. For each of those families the Picard-Fuchs equation has a unique Frobenius basis of solutions near 0:

$$y_0(t) = f_0(t), \tag{45}$$

$$y_1(t) = \log(t) \cdot f_0(t) + f_1(t), \tag{46}$$

with $f_0(t) \in \mathbb{Z}[[t]]$, $f_1(t) \in t\mathbb{Z}[[t]]$. Let $Y(t)$ denote the fundamental solution matrix:

$$Y(t) = \begin{pmatrix} y_0(t) & \theta(y_0(t)) \\ y_1(t) & \theta(y_1(t)) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \log(t) & 1 \end{pmatrix} \cdot \begin{pmatrix} f_0(t) & \theta(f_0(t)) \\ f_1(t) & \theta(f_1(t)) \end{pmatrix}. \tag{47}$$

The matrix $U(t)$ from above has the form

$$U(t) = Y(t^p)^{-1} \begin{pmatrix} 1 & 0 \\ pb & p \end{pmatrix} Y(t). \tag{48}$$

It turns out experimentally that the correct value of b, such that the entries $U(t)$ are $p-$adic power series with a nice property, is $b = 0$ for all the above mentioned cases:

$$U(0) = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}. \tag{49}$$

Experimentally in many examples it can be observed that the power series entries have the following nice shape:

$$\begin{aligned} \Delta(t)^{p\cdot\delta_0} \cdot U(t) &= V_0(t) \mod p, \\ \Delta(t)^{p\cdot\delta_1} \cdot U(t) &= V_1(t) \mod p^2, \\ \Delta(t)^{p\cdot\delta_2} \cdot U(t) &= V_2(t) \mod p^3 \end{aligned} \tag{50}$$

etc where the $V_N \in Mat(2 \times 2, \mathbb{Z}_p[t])$ of degree d_N. One gets a p-adic expansion

$$U(t) = U_0(t) + p \cdot U_1(t) + p^2 \cdot \frac{U_2(t)}{\Delta(t)^{p\cdot\delta_2}} + p^3 \cdot \frac{U_3(t)}{\Delta(t)^{p\cdot\delta_3}} + \dots. \tag{51}$$

In our examples we try to find out experimentally the δ_N such that

$$\Delta(t)^{p\cdot\delta_{N-1}} \cdot U(t) \mod p^N \tag{52}$$

is a matrix with polynomial entries with polynomials of degree d_N.

Example: Legendre family

We consider the scaled hypergeometric operator

$$\theta^2 - 16t\left(\theta + \frac{1}{2}\right)^2, \tag{53}$$

which has discriminant $\Delta(t) = t(1 - 16t)$ and Riemann symbol

$$\left\{ \begin{array}{ccc} 0 & 1/16 & \infty \\ \hline 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \end{array} \right\} \tag{54}$$

We get for (51) the degrees $d_N = \lfloor (p-1)/2 \rfloor$ and exponents $\delta_N = N$.

Example: Apery family

We consider the scaled hypergeometric operator

$$\theta^2 - t\left(11\theta^2 + 11\theta + 3\right) - t^2\left(\theta + 1\right)^2, \tag{55}$$

which has discriminant $\Delta(t) = t(1 - 11t - t^2)$ and Riemann symbol

$$\left\{ \begin{array}{cccc} 0 & \frac{-11+5\sqrt{5}}{2} & \frac{-11-5\sqrt{5}}{2} & \infty \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right\} \tag{56}$$

We get for (51) the degrees $d_N = p - 1$ and exponents $\delta_N = 2N$.

5.5. **The platonic cases.** Surprisingly, the Ansatz 49 for $U(0)$ does not work for the platonic operators. Taking the diagonal shape does not lead to converging power series mod p^2.

It turns out that one needs to return to the more general Ansatz for $U(0)$:

$$U_p(0) = \begin{pmatrix} 1 & 0 \\ x_p & p \end{pmatrix}. \tag{57}$$

We were able to identify the unknown terms x_p in the U-matrices of our Herfurtner operators: they are all log-terms of the form

$$x_p = (p-1)\frac{1}{n}\log_p(c), \tag{58}$$

where n comes form the fiber type I_n at 0. The c that we detected are summarized in the following table:

Operator 7	c	Local exponents at ∞	δ_N	d_N
$\mathbb{T}(3)$	2^3	$2/3, 4/3$	$\lfloor \frac{2}{3}(p-1) \rfloor$	$2N$
$\mathbb{T}(2)$	3			
$\mathbb{O}(4)$	3^2	$3/4, 5/4$	$\lfloor \frac{3}{4}(p-1) \rfloor$	
$\mathbb{O}(3)$	2^2			
$\mathbb{O}(2)$	3^4			
$\mathbb{I}(5)$	2^3	$4/5, 6/5$	$\lfloor \frac{4}{5}(p-1) \rfloor$	
$\mathbb{I}(3)$	$2^5 \cdot 5^2$			
$\mathbb{I}(2)$	5^3			

With those entries for x_p in the matrices $U(0)$ the entries of $U(t)$ converge nicely as in (51).

Proposition Let $\mathcal{E} \to \mathbb{P}^1$ ... rational??? elliptic surface with multiplicative fiber type I_n at 0 and with j-invariant $j(t)$ and $y_0(t), y_1(t)$ the Frobenius basis of the Picard-Fuchs operator. The x_p in 49 are given by

$$x_p = (p-1)\frac{1}{n}\log_p(c) \tag{59}$$

with

$$c = \left(\frac{1}{j(t)} \Big/ \exp\left(y_1(t)/y_0(t)\right)\right)\Bigg|_{t=0}. \tag{60}$$

6. Monodromy calculations

The computations of the monodromy matrices of the differential equations with respect to the Frobenius basis lead to interesting observations. For the calculations we used the ORE-algebra package in SAGE.

There are three different configurations of the singular fibers which can be seen in Figure 11. With s_0 we denote the singular point 0, with s_1 the singular point that is nearest to 0 in the real case, and the complex point in the upper half plane in the complex conjugate case, and with s_2 the remaining singularity $\neq \infty$. The choices of the base point and the paths in each configuration can be seen in Figure 11.

Example: Icosahedral operator As an example of the monodromy matrices we give the icosahedral operator $\mathbb{I}(5)$.

The singular fibers are I_5 at 0, I_2 at $\frac{1}{16}$, I_3 at $\frac{2}{27}$ and II at ∞. For the monodromy calculations we first choose paths around the finite singular points, which are visible in Figure 10.

With those paths with the Frobenius basis at 0 we get the following monodromy matrices:

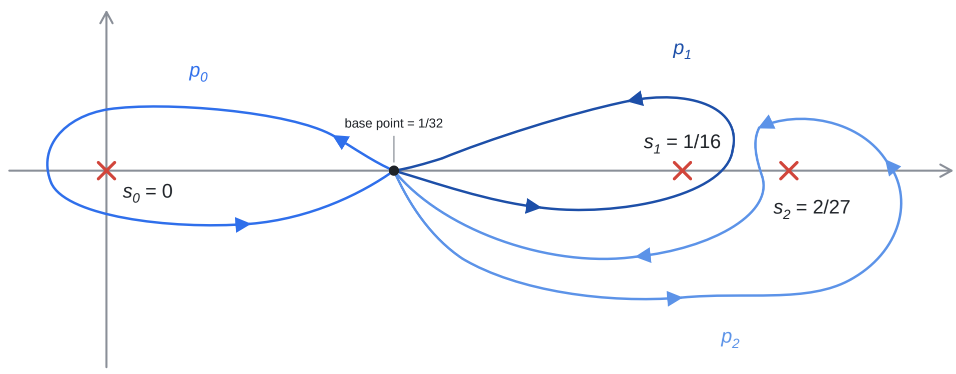


FIGURE 10. Paths around the singular points of $\mathbb{I}(5)$

$$M_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \; M_1 = \begin{pmatrix} 1 + \dfrac{2\log(2^3)}{2\pi i} & -10 \\ \dfrac{2}{5}\left(\dfrac{\log(2^3)}{2\pi i}\right)^2 & 1 - \dfrac{2\log(2^3)}{2\pi i} \end{pmatrix},$$

(61)

$$M_2 = \begin{pmatrix} 4 + \dfrac{3\log(2^3)}{2\pi i} & -15 \\ \dfrac{3}{5}\left(1 + \dfrac{\log(2^3)}{2\pi i}\right)^2 & -2 - \dfrac{3\log(2^3)}{2\pi i} \end{pmatrix}.$$

The matrix, that transforms the above matrices to integral forms, is:

$$T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i} \cdot \log(2^3) & 5 \end{pmatrix}$$

The integer matrices after transformation $T \cdot M \cdot T^{-1}$ are then:

$$\tilde{M}_0 = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}, \quad \tilde{M}_{1/16} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad \tilde{M}_{2/27} = \begin{pmatrix} 4 & -3 \\ 3 & -2 \end{pmatrix}.$$

Those give us the monodromy group $SL_2(\mathbb{Z})$.

We see that the non-rational entry appearing in the transformation matrix is equal to the first non-zero coefficient c of $\frac{1}{j(t)}$ (60) of the corresponding family.

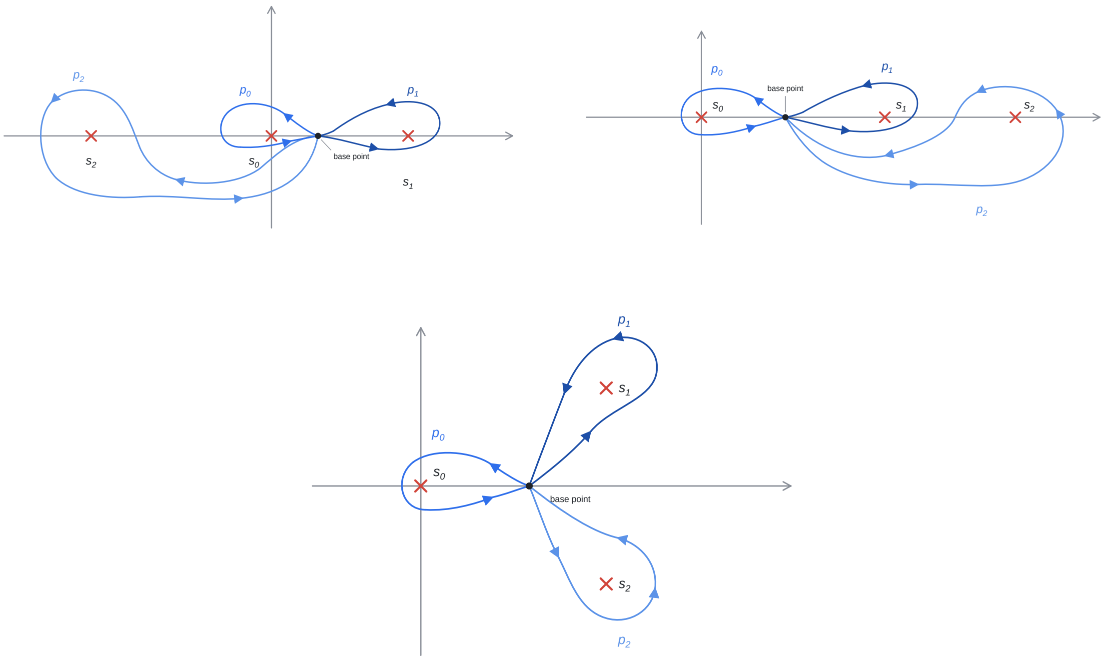


Figure 11. Three different configurations of the singular points with counterclockwise generators of $\pi_1(\mathbb{P}^1 \setminus \{s_0, s_1, s_2, \infty\})$.

7. Outlook

Starting from the reflection groups in three-space we were led to an interesting class of elliptic surfaces. There are several ways one can seek generalisations. We briefly mention a few further directions.

7.1. **Other sections.** In section 1.6 we restricted the elliptic threefold to the lines $P = r$, where r is a non-zero constant. A different choice of the lines in the $P - Q$ plane lead to different elliptic surfaces. Table 8 shows all possible configurations of lines. Here r denotes a constant and t the parameter of the family.

7.2. **Other quotients.** Previous article in this issue One may wonder if there are other rings of invariants that lead to elliptic fibrations. It turns out there are quite a number of further cases that deserve closer study.

Specialization	**Tetrahedron**	**Octahedron**	**Icosahedron**
$P=r,\ Q=t, r\neq 0$	$\mathrm{I}_2\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{IV}$	$\mathrm{I}_4\ \mathrm{I}_2\ \mathrm{I}_3\ \mathrm{III}$	$\mathrm{I}_2\ \mathrm{I}_5\ \mathrm{I}_3\ \mathrm{II}$
$P=0,\ Q=t$	$\mathrm{IV}^*\ \mathrm{IV}$	$\mathrm{III}^*\ \mathrm{III}$	$\mathrm{II}^*\ \mathrm{II}$
$P=t,\ Q=r, r\neq 0$	$\mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_3$	$\mathrm{I}_2\ \mathrm{I}_2\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_8^*$	$\mathrm{I}_5\ \mathrm{I}_5\ \mathrm{I}_5\ \mathrm{I}_3$ $\mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_6^*$
$P=t,\ Q=0$	Singular generic fiber		
$P=mt+r,\ Q=t,$ $m,r\neq 0$, generic	$m^2r\neq 4$ $\mathrm{I}_2\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_1$	$mr\notin\left\{1,\frac{3}{4}\right\}$ $\mathrm{I}_4\ \mathrm{I}_2\ \mathrm{I}_2\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_4^*$	$mr^2\notin\left\{\frac{4}{27},-\frac{4}{5}\right\}$ $\mathrm{I}_2\ \mathrm{I}_5\ \mathrm{I}_5\ \mathrm{I}_5$ $\mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_4^*$
$P=mt+r,\ Q=t,$ tangent	$m^2r=4$ $\mathrm{I}_2\ \mathrm{I}_6\ \mathrm{I}_3\ \mathrm{I}_1$	$mr=1$ $\mathrm{I}_4\ \mathrm{I}_4\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_4^*$ $mr=\frac{3}{4}$ $\mathrm{I}_4\ \mathrm{I}_2\ \mathrm{I}_2\ \mathrm{I}_6\ \mathrm{I}_4^*$	$mr^2=\frac{4}{27}$ $\mathrm{I}_2\ \mathrm{I}_{10}\ \mathrm{I}_5\ \mathrm{I}_3$ $\mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_4^*$ $mr^2=-\frac{4}{5}$ $\mathrm{I}_2\ \mathrm{I}_5\ \mathrm{I}_5\ \mathrm{I}_5$ $\mathrm{I}_6\ \mathrm{I}_3\ \mathrm{I}_4^*$
$P=mt,\ Q=t,$ $m\neq 0$	$\mathrm{IV}^*\ \mathrm{I}_3\ \mathrm{I}_1$	$\mathrm{III}^*\ \mathrm{I}_2\ \mathrm{I}_3\ \mathrm{I}_4^*$	$\mathrm{II}^*\ \mathrm{I}_5\ \mathrm{I}_5\ \mathrm{I}_3\ \mathrm{I}_3\ \mathrm{I}_4^*$

TABLE 8. Kodaira fiber configurations of the tetrahedral, octahedral, and icosahedral families

One can look at invariant rings of rotation subgroups of other reflection groups. We mention the ring of invariants for the Valentiner group, appearing as number XXX in the the Shephard-Todd list. The ring of invariants is of the form

$$XXX$$

and leads to elliptic surface XXX.

But also invariant rings of more general algebraic groups may be considered. For example, for the action of $Sl_2(\mathbb{C})$ acting on binary forms of degree five, it is a classical fact that its ring of invariants of binary quintics is generated by invariants I_4, I_8, I_{12}, I_{18}, with a single relation, [17] In the paper [1] simpler invariants J, K, L, H are defined, among which there exists the relation (eq. (32) in [1]):

$$16H^2 = -432L^3 - 72L^2KJ + 8LK^3 - 2LK^2J^2 + L^2J^3 + K^4J.$$

For fixed value of $J \neq 0$ we obtain over the K-line the cousin $135III$ of the octahedral case as elliptic curves written in the variables L and H.

7.3. **Some further operators.** In the list of Herfürtner, there are some further cases where the Picard-Fuchs operator has degree 2. For example, there are 13 cases with two semi-stable and two reduced non I_n-fibres. [2]

By bringing these non semi-simple fibres to 0 and ∞ one obtains an operator of degree 2, but of course these operators have finite local monodromy at 0 and ∞ but lead to two term recursion for the coefficients.

7.4. **Apery-like limits.** The Picard–Fuchs operator

$$\mathcal{L} = P_0(\theta) + t\,P_1(\theta) + t^2\,P_2(\theta),$$

yields the two-term recurrence

$$P_0(n)\,a_n + P_1(n-1)\,a_{n-1} + P_2(n-2)\,a_{n-2} = 0.$$

Let $(a_n)_n$ be the solution with $a_{-1} = 0$, $a_0 = 1$ (the normalised period integral, whose coefficients are the integers listed in Table 10), and let $(b_n)_n$ be the second solution of the same recurrence with $b_0 = 0$, $b_1 = 1$. In analogy with Apéry's proof of the irrationality of $\zeta(3)$ and $\zeta(2)$ one is led to consider the limit

$$L := \lim_{n\to\infty} \frac{b_n}{a_n}.$$

For example for the Apéry operator $\theta^2 - t(11\theta^2 + 11\theta + 3) - t^2(\theta+1)^2$ this recovers $L = \zeta(2)/5$.

Table 9 collects, for each platonic operator, the limit. In some cases the number was identified and turns out to be a logarithmic term.

[2]The cases are $II\,II\,7\,1$, $II\,II\,6\,2$, $II\,II\,4\,4$, $II\,III\,6\,1$, $II\,III\,5\,2$, $II\,III\,4\,3$, $II\,IV\,5\,1$, $II\,VI\,4\,2$, $III\,III\,5\,1$, $III\,III\,3,3$, $III\,IV\,4\,1$, $III\,VI\,3\,2$, $IV\,IV\,2\,2$

Operator	Decimal approximation of $L = \lim_{n\to\infty} \frac{b_n}{a_n}$	Exact value
$\mathbb{T}(3) = 23\underline{3}\mathrm{IV}$	$0.15403270679\ldots$	$\frac{2}{3^2}\log 2$
$\mathbb{T}(2) = \underline{2}33\mathrm{IV}$	does not exist (HG pullback)	—
$\mathbb{O}(4) = 23\underline{4}\mathrm{III}$	$0.20598980413\ldots$	$\frac{3}{2^4}\log 3$
$\mathbb{O}(3) = 2\underline{3}4\mathrm{III}$	$0.23104906019\ldots$	$\frac{1}{3}\log 2$
$\mathbb{O}(2) = \underline{2}34\mathrm{III}$	$0.04872722313\ldots$	open
$\mathbb{I}(5) = 23\underline{5}\mathrm{II}$	$0.27725887222\ldots$	$\frac{2}{5}\log 2$
$\mathbb{I}(3) = 2\underline{3}5\mathrm{II}$	$0.04132287987\ldots$	$\frac{5}{3^3}\log\left(\frac{5}{2^2}\right)$
$\mathbb{I}(2) = \underline{2}35\mathrm{II}$	$0.03746947912\ldots$	open

TABLE 9. Laurent polynomials and Apery-like limits of the Platonic operators

APPENDIX A. GOURSAT INVARIANTS

The invariant P always has the form

$$P = x^2 + y^2 + z^2.$$

Tetrahedron.

$$\begin{aligned} Q &= x^2y^2 + y^2z^2 + x^2z^2, \\ R &= xyz, \\ S &= (x^2 - y^2)(y^2 - z^2)(z^2 - x^2). \end{aligned}$$

Octahedron.

$$\begin{aligned} Q &= x^2y^2 + y^2z^2 + x^2z^2, \\ R &= x^2y^2z^2, \\ S &= xyz(x^2 - y^2)(y^2 - z^2)(z^2 - x^2). \end{aligned}$$

Icosahedron.

$$\begin{aligned} Q &= z(-5z^3(x^2+y^2) + 5z(x^2+y^2)^2 - 2x(x^4 - 10x^2y^2 + 5y^4) + z^5), \\ R &= (4x^2 + 6xz + z^2)(x^4 + 8x^3z - 10x^2y^2 + 14x^2z^2 - 8xz^3 + 5y^4 - 10y^2z^2 + z^4) \\ &\quad (x^4 - 2x^3z - 10x^2y^2 - x^2z^2 + 30xy^2z + 2xz^3 + 5y^4 - 25y^2z^2 + z^4), \\ S &= 8y(5x^4 - 10x^2y^2 + y^4)(x^2 - xz - z^2) \\ &\quad (x^4 - 10x^2y^2 + 5y^4 - 12x^3z + 20xy^2z + 44x^2z^2 - 20y^2z^2 - 48xz^3 + 16z^4) \\ &\quad (x^4 - 10x^2y^2 + 5y^4 + 8x^3z - 40xy^2z + 24x^2z^2 - 40y^2z^2 + 32xz^3 + 16z^4). \end{aligned}$$

Appendix B. Surfaces with three I_n-fibers and one II, III or IV-fiber

B.1. **Picard-Fuchs operators.** By multiplying the solutions with an elementary factor, the solutions of any differential equation with four regular singular singularities can be reduced to the Heun equation

$$\frac{d^2y}{dx^2} + (\frac{\gamma}{x} + \frac{\delta}{x-1} + \frac{\epsilon}{x-t})\frac{dy}{dx} + \frac{\alpha\beta x - q}{x(x-1)(x-t)}y = 0$$

has Riemann symbol

$$\left\{ \begin{array}{cccc} 0 & 1 & t & \infty \\ \hline 0 & 0 & 0 & \alpha \\ 1-\gamma & 1-\delta & 1-\epsilon & \beta \end{array} \right\}$$

and accessory parameter q. The Picard-Fuchs operators which arise from elliptic surfaces from Herfurthners list are all of Heun type and can be found in [47]. Here we collect the basic data for the platonic family, i.e. those elliptic surfaces with four *reduced* fibres, three of which are semi-stable. The Picard-Fuchs operators that annihilate the normalised period integrals are of degree 2, hence lead to two term recursions. The operators can be written as

$$a\theta^2 - t(b\theta^2 + b\theta + \lambda) + ct^2(\theta + \alpha_1)(\theta + \alpha_2), \tag{62}$$

where a, b, c are integers and the local exponents are

$$(\alpha_1, \alpha_2) = \begin{cases} \left(\frac{5}{6}, \frac{7}{6}\right), & \text{for } II, \\ \left(\frac{3}{4}, \frac{5}{4}\right), & \text{for } III, \\ \left(\frac{2}{3}, \frac{4}{3}\right), & \text{for } IV. \end{cases}$$

The Riemann symbol for such an operator has the form

$$\left\{ \begin{array}{cccc} 0 & * & * & \infty \\ \hline 0 & 0 & 0 & \alpha_1 \\ 0 & 0 & 0 & \alpha_2 \end{array} \right\} \tag{63}$$

so we are dealing with a *Heun equation* with a special exponent structure,

$$\gamma = \delta = \epsilon = 1.$$

Around 0 and the two singularities denoted by $*$, there is a holomorphic solution ϕ and a logarithmic solution ψ, so that the local monodromy matrices are conjugate to the Jordan block

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{64}$$

At ∞, the monodromy is conjugate to the matrix

$$\begin{pmatrix} e^{2\pi i \alpha_1} & 0 \\ 0 & e^{2\pi i \alpha_2} \end{pmatrix} \tag{65}$$

hence is of finite order $= e$. Indeed, pulling back the elliptic surface near ∞ by an e-fold cover, the fibre at ∞ is a smooth elliptic curve with an automorphism of order e.

Table 10 collects some information on these operators: we record the Weierstrass covariants g_2 and g_3 for the elliptic surface, the critically scaled Picard-Fuchs operator and the location of the singular point, the first few coefficients of the integer sequence, a Laurent polynomial generating that sequence, the Apéry constant L (if available), the integral monodromy matrices, the transformation T connecting Frobenius and integral basis and the constant c.

TABLE 10. List of the Herfurtner operators

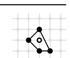
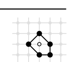
Notation	$g_2(t), g_3(t)$ for the Weierstrass form Picard-Fuchs operator Critically scaled $(a_n)_n$ Laurent polynomial Decimal approximation of $L = \lim_{n\to\infty} b_n/a_n$ and exact value (if identified) Monodromy matrices M_0, M_{s_1}, M_{s_2}	Configuration of Singular fibers $0, s_1, s_2, \infty$	First coefficient c of $\frac{1}{j(t)}$ Transformation matrix	Monodromy group
$11\underline{8}II$	$g_2 = -64t^3 + 20t^2 - 2t + 1/12$ $g_3 = 64t^5 - 80t^4 + 64/3t^3 - 8/3t^2 + 1/6t - 1/216$ $3\theta^2 - t \cdot (56\theta^2 + 56\theta + 18) + t^2 \cdot 432\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 6, 30, 12, -2250, -35244, -329364, \ldots$ $F = \frac{3x+4x^2-xy^2+4y+6xy}{xy}$  L : does not exist $M_{s_1} = \begin{pmatrix} -1 & -1 \\ 4 & 3 \end{pmatrix}, M_{s_2} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, M_0 = \begin{pmatrix} 1 & 0 \\ 8 & 1 \end{pmatrix}$	$I_1 \; I_1 \; I_8 \; II$ $\frac{7\pm 4i\sqrt{2}}{108} \; 0 \; \infty$	$-2^8 \cdot 3$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-2^8 \cdot 3) & 8 \end{pmatrix}$	$SL_2(\mathbb{Z})$
$12\underline{7}II$	$g_2 = -28t^3 + 14t^2 - 2t + 1/12$ $g_3 = 16t^5 - 29t^4 + 37/3t^3 - 13/6t^2 + 1/6t - 1/216$ $6\theta^2 - t \cdot (113\theta^2 + 113\theta + 36) + t^2 \cdot 432\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 6, 48, 444, 4500, 48456, 543816, \ldots$ $F = \frac{3x+2x^2+3x^2y+xy^2+y+6xy}{xy}$  $L = 0.27279178641\cdots = \frac{1}{7}\log\left(\frac{3^3}{2^2}\right)$ $M_{s_1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, M_{s_2} = \begin{pmatrix} 5 & -2 \\ 8 & -3 \end{pmatrix}, M_0 = \begin{pmatrix} 1 & 0 \\ 7 & 1 \end{pmatrix}$	$I_1 \; I_2 \; I_7 \; II$ $\frac{2}{27} \; \frac{3}{16} \; 0 \; \infty$	$2 \cdot 3^2$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2 \cdot 3^2) & 7 \end{pmatrix}$	$SL_2(\mathbb{Z})$

Table 10 – continued

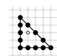
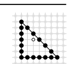
$1\underline{2}7II$	Substitute $t \mapsto -49t + 3/16$ $3\theta^2 - t \cdot (2080\theta^2 + 2080\theta + 684)$ $+t^2 \cdot 338688 \left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 228, 64596, 20133456, 6666855300, \ldots$ $F = \frac{N}{xy^2}$, where  $N = -729 + 36y + 186y^2 + 4y^3 - 9y^4 - 324x$ $- 100xy + 228xy^2 - 12xy^3 + 450x^2 - 36x^2y$ $+ 50x^2y^2 + 108x^3 + 36x^3y - 81x^4$ $L = 0.00903720141 \cdots = \frac{1}{16\sqrt{7}} \log\left(\frac{29 + 4\sqrt{7}}{3^3}\right)$ $M_{s_1} = \left(\begin{smallmatrix} -1 & -4 \\ 1 & 3 \end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 1 & -7 \\ 0 & 1 \end{smallmatrix}\right)$	$I_1 \; I_2 \; I_7 \; II$ $\frac{1}{432} \; 0 \; \frac{3}{784} \; \infty$	-3^7 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-3^7) & 2 \end{pmatrix}$	
$\underline{1}27II$	Substitute $t \mapsto -49t + 2/27$ $2\theta^2 - t \cdot (459\theta^2 + 459\theta + 132) - t^2 \cdot 571536 \left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 66, 78120, 20849556, 16193337300, \ldots$ $F = \frac{N}{x^2y^3}$, where  $N = 1024 + 768y + 528y^2 + 400y^3 + 132y^4 + 48y^5$ $+ 16y^6 + 768x + 480xy + 264xy^2 + 150xy^3$ $+ 33xy^4 + 6xy^5 + 3264x^2 + 1632x^2y$ $+ 333x^2y^2 + 66x^2y^3 - 57x^2y^4 + 1552x^3$ $+ 582x^3y + 75x^3y^2 - 76x^3y^3 + 3264x^4$ $+ 816x^4y - 228x^4y^2 + 768x^5 + 96x^5y$ $+ 1024x^6$ $L = 0.00416061797\ldots$ (open) $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 5 & -8 \\ 2 & -3 \end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix} 1 & -7 \\ 0 & 1 \end{smallmatrix}\right)$	$I_1 \; I_2 \; I_7 \; II$ $0 \; -\frac{1}{432} \; \frac{2}{1323} \; \infty$	2^7 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^7) & 1 \end{pmatrix}$	

Table 10 – continued

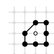
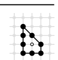
$14\underline{5}II$	$g_2 = 640t^3 + 140t^2 - 10t + 1/12$ $g_3 = -1024t^5 - 3200t^4 + 440/3t^3 - 110/3t^2$ $+5/6t - 1/216$ $3\theta^2 - t \cdot (328\theta^2 + 328\theta + 90) + t^2 \cdot 432\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 30, 1830, 137580, 11391750, 997411140, \ldots$ $F = \frac{9x^2y^2+12x^2y+4x^2+16xy^2+30xy+12x+16y+9}{xy}$  $L = 0.03397980736\cdots = \frac{1}{5}\log\left(\frac{2^5}{3^3}\right)$ $M_{s_1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, M_{s_2} = \begin{pmatrix} 9 & -4 \\ 16 & -7 \end{pmatrix}, M_0 = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$	$I_1 \ I_4 \ I_5 \ II$ $\frac{1}{108} \ \frac{3}{4} \ 0 \ \infty$	$2^{10} \cdot 3^4$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^{10}\cdot 3^4) & 5 \end{pmatrix}$	$SL_2(\mathbb{Z})$
$1\underline{4}5II$	$g_2 = -40960t^3 + 1580t^2 - 20t + 1/12$ $g_3 = 4194304t^5 - 450560t^4 + 45640/3t^3 - 695/3t^2$ $+5/3t - 1/216$ $15\theta^2 - t \cdot (2576\theta^2 + 2576\theta + 900)$ $+t^2 \cdot 110592\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 60, 4260, 320880, 24920100, 1972024560, \ldots$ $F = \frac{3x^2y-x^2+22xy^2+60xy+6x-45y^3-51y^2+49y-9}{xy}$  $L = 0.06587495912\cdots = \frac{5}{2^7}\log\left(\frac{3^3}{5}\right)$ $M_{s_1} = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}, M_0 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}, M_{s_2} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$	$I_1 \ I_4 \ I_5 \ II$ $\frac{5}{432} \ 0 \ \frac{3}{256} \ \infty$	$-3^5 \cdot 5$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-3^5\cdot 5) & 4 \end{pmatrix}$	

Table 10 – continued

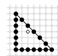
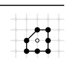
$\underline{1}45II$	Substitute $t \mapsto -64t + 1/108$ $5\theta^2 - t \cdot (341280\theta^2 + 341280\theta + 9300)$ $-t^2 \cdot 2985984\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 1860, 7357860, 35492084880, 188281083032100, \ldots$ $F = \frac{N}{x^2y^3}$, where  $N = -1953125 + 1218750y - 433875y^2 + 107300y^3 - 17355y^4 + 1950y^5 - 125y^6 + 1875000x - 705000xy + 139440xy^2 - 17808xy^3 + 24xy^4 + 120xy^5 - 834375x^2 + 174300x^2y - 17622x^2y^2 + 1860x^2y^3 - 3x^2y^4 + 214000x^3 - 17520x^3y + 48x^3y^2 + 16x^3y^3 - 33375x^4 + 30x^4y - 3x^4y^2 + 3000x^5 + 120x^5y - 125x^6$ $L = 0.00052717285\ldots$ (open) $M_0 = \left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}9 & -16\\ 4 & -7\end{pmatrix}$, $M_{s_1} = \left(\begin{smallmatrix}1 & -5\\ 0 & 1\end{smallmatrix}\right)$	$I_1 \quad I_4 \quad I_5 \quad II$ $0 \quad -\frac{5}{432} \quad \frac{1}{6912} \quad \infty$	5^4 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(5^4) & 1\end{pmatrix}$	
$23\underline{5}II$	$g_2 = -180t^3 + 130/3t^2 - 10/3t + 1/12$ $g_3 = 432t^5 - 325t^4 + 1855/27t^3$ $-115/18t^2 + 5/18t - 1/216$ $2\theta^2 - t \cdot (59\theta^2 + 59\theta + 20) + t^2 \cdot 432\left(\theta + \frac{7}{6}\right)\left(\theta + \frac{5}{6}\right)$ $a_n = 1, 10, 120, 1540, 20500, 279480, 3876600, \ldots$ $F = \frac{2-3x-2x^2+3x^2y+2x^2y^2-xy^2+y+10xy}{xy}$  $L = 0.27725887222\cdots = \frac{2}{5}\log 2$ $M_{s_1} = \left(\begin{smallmatrix}1 & -2\\ 0 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}4 & -3\\ 3 & -2\end{pmatrix}$, $M_0 = \left(\begin{smallmatrix}1 & 0\\ 5 & 1\end{smallmatrix}\right)$	$I_2 \quad I_3 \quad I_5 \quad II$ $\frac{1}{16} \quad \frac{2}{27} \quad 0 \quad \infty$	2^3 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(2^3) & 5\end{pmatrix}$	$SL_2(\mathbb{Z})$

Table 10 – continued

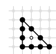
$2\underline{3}5II$	$g_2 = 14580t^3 + 270t^2 - 10t + 1/12$ $g_3 = -314928t^5 - 120285t^4 + 2835t^3 - 95/2t^2$ $+5/6t - 1/216$ $10\theta^2 - t \cdot (999\theta^2 + 999\theta + 300)$ $+t^2 \cdot 11664\,(\theta + \frac{7}{6})\,(\theta + \frac{5}{6})$ $a_n = 1, 30, 1440, 85260, 5606100, 391231080, \ldots$ $F = \frac{2x^3-3x^2y-12xy^2+20y^3-3x^2+30xy-21y^2-12x-21y+20}{xy}$  $L = 0.04132287987\cdots = \frac{5}{3^3}\log\left(\frac{5}{2^2}\right)$ $M_{s_1} = \left(\begin{smallmatrix}1 & -2\\ 0 & 1\end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix}1 & 0\\ 3 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}6 & -5\\ 5 & -4\end{pmatrix}$	I_2 I_3 I_5 II $\frac{5}{432}$ 0 $\frac{2}{27}$ ∞	$2^5 \cdot 5^2$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^5 \cdot 5^2) & 3 \end{pmatrix}$	
$\underline{2}35II$	Substitute $t \mapsto t + 1/16$ $5\theta^2 - t \cdot (352\theta^2 + 352\theta + 100)$ $-t^2 \cdot 6912\,(\theta + \frac{7}{6})\,(\theta + \frac{5}{6})$ $a_n = 1, 20, 1140, 68240, 4572100, 321427920, \ldots$ F : does not exist $L = 0.03746947912\ldots$ (open) $M_0 = \left(\begin{smallmatrix}1 & 0\\ 2 & 1\end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix}1 & -3\\ 0 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}6 & -5\\ 5 & -4\end{pmatrix}$	I_2 I_3 I_5 II 0 $\frac{5}{432}$ $-\frac{1}{16}$ ∞	5^3 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(5^3) & 2 \end{pmatrix}$	

Table 10 – continued

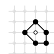
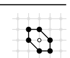
$11\underline{7}III$	$g_2 = -4t^3 + 10/3t^2 - 2/3t + 1/12$ $g_3 = -7/3t^4 + 35/27t^3 - 7/18t^2 + 1/18t - 1/216$ $2\theta^2 - t \cdot (13\theta^2 + 13\theta + 4) + t^2 \cdot 64\left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 2, 0, -28, -140, -168, 2184, 16200, \ldots$ $F = \frac{(x+y)(xy-2x-1)}{xy}$  L : does not exist $M_{s_1} = \left(\begin{smallmatrix} -1 & -1 \\ 4 & 3 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 7 & 1 \end{smallmatrix}\right)$	I_1 I_1 I_7 III $\frac{13 \pm i\sqrt{7}}{128}$ 0 ∞	-2 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-2) & 7 \end{pmatrix}$	$\mathrm{SL}_2(\mathbb{Z})$
$12\underline{6}III$ $\left(\overset{\text{isog.}}{\simeq} 2\underline{3}4III\right)$	$g_2 = -16t^3 + 12t^2 - 2t + 1/12$ $g_3 = -16t^4 + 28/3t^3 - 2t^2 + 1/6t - 1/216$ $\theta^2 - t \cdot (20\theta^2 + 20\theta + 6) + t^2 \cdot 64\left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 6, 54, 588, 7110, 91476, 1224636, \ldots$ $F = \frac{(2x+y+1)(xy+x+2y)}{xy}$  L : see $2\underline{3}4III$ $M_{s_1} = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 5 & -2 \\ 8 & -3 \end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 6 & 1 \end{smallmatrix}\right)$	I_1 I_2 I_6 III $\frac{1}{16}$ $\frac{1}{4}$ 0 ∞	2^4 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^4) & 6 \end{pmatrix}$	$\Gamma_0(2)$

Table 10 – continued

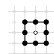
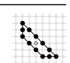
$1\underline{2}6III$ $\left(\overset{\text{isog.}}{\simeq} 23\underline{4}III\right)$	Substitute $t \mapsto -4t + 1/4$ $3\theta^2 - t \cdot (112\theta^2 + 112\theta + 36)$ $+t^2 \cdot 1024 \left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 12, 180, 2928, 49860, 875952, \ldots$ $F = \frac{(xy-x+3y+1)(xy+3x-y+1)}{xy}$ (as for $23\underline{4}III$)  L : see $23\underline{4}III$ $M_{s_1} = \left(\begin{smallmatrix} -1 & -4 \\ 1 & 3 \end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 1 & -6 \\ 0 & 1 \end{smallmatrix}\right)$	$I_1\ I_2\ I_6\ III$ $\frac{3}{64}\ 0\ \frac{1}{16}\ \infty$	-3 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-3) & 2 \end{pmatrix}$	
$\underline{1}26III$ $\left(\overset{\text{isog.}}{\simeq} \underline{2}34III\right)$	Substitute $t \mapsto -4t + 1/16$ $3\theta^2 - t \cdot (128\theta^2 + 128\theta + 36)$ $-t^2 \cdot 4096 \left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 12, 612, 25392, 1298340, 67041072, \ldots$ $F = \frac{N}{x^2y^2}$, where  $N = -3y^3 + 18y^4 - 27y^5 - xy^2 - 8xy^3 + 33xy^4$ $\quad - x^2y + 12x^2y^2 - 22x^2y^3 - 3x^3 - 8x^3y$ $\quad - 22x^3y^2 + 18x^4 + 33x^4y - 27x^5$ L : see $\underline{2}34III$ $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 5 & -8 \\ 2 & -3 \end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix} 1 & -6 \\ 0 & 1 \end{smallmatrix}\right)$	$I_1\ I_2\ I_6\ III$ $0\ -\frac{3}{64}\ \frac{1}{64}\ \infty$	3^2 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(3^2) & 1 \end{pmatrix}$	

Table 10 – continued

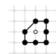
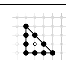
$13\underline{5}III$	$g_2 = 4t^3 + 6t^2 - 2t + 1/12$ $g_3 = -5t^4 + 5/3t^3 - 3/2t^2 + 1/6t - 1/216$ $6\theta^2 - t \cdot (131\theta^2 + 131\theta + 36) + t^2 \cdot 64 \left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 6, 72, 1068, 17460, 301896, 5414136, \ldots$ $F = \frac{1+2x+x^2+4x^2y+4x^2y^2+3xy^2+2y+6xy}{xy}$  $L = 0.17260924347\cdots = \frac{3}{5}\log\left(\frac{2^2}{3}\right)$ $M_{s_1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 7 & -3 \\ 12 & -5 \end{pmatrix}$, $M_0 = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$	I_1 I_3 I_5 III $\frac{3}{64}$ 2 0 ∞	$2^3 \cdot 3$ $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^3 \cdot 3) & 5 \end{pmatrix}$	$\mathrm{SL}_2(\mathbb{Z})$
$1\underline{3}5III$	$g_2 = -12500t^3 + 750t^2 - 14t + 1/12$ $g_3 = -78125t^4 + 14375/3t^3 - 223/2t^2$ $+7/6t - 1/216$ $2\theta^2 - t \cdot (253\theta^2 + 253\theta + 84)$ $+t^2 \cdot 8000 \left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 42, 2160, 118740, 6750900, 391745592, \ldots$ $F = \frac{(x+y-2)(4x^2-17xy+4x+4y^2+4y+1)}{xy}$  $L = 0.09241962407\cdots = \frac{2}{15}\log 2$ $M_0 = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$, $M_{s_1} = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$	I_1 I_3 I_5 III $\frac{1}{64}$ 0 $\frac{2}{125}$ ∞	-2^5 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(-2^5) & 3 \end{pmatrix}$	

Table 10 – continued

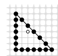
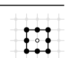
$\underline{1}35III$	Substitute $t \mapsto -125t + 3/64$ $3\theta^2 - t \cdot (78080\theta^2 + 78080\theta + 2124)$ $-t^2 \cdot 512000\,(\theta + \frac{5}{4})\,(\theta + \frac{3}{4})$ $a_n = 1, 708, 1086660, 2023823760, 4143653100900, \ldots$ $F = \frac{N}{x^2y^3}$, where  $N = 14348907 - 2243862y - 161595y^2 + 58860y^3 - 1995y^4 - 342y^5 + 27y^6 + 3188646x - 135594xy - 14724xy^2 + 1676xy^3 - 34xy^4 + 30xy^5 + 767637x^2 - 38556x^2y - 1794x^2y^2 + 708x^2y^3 + 5x^2y^4 + 84564x^3 + 1836x^3y - 4x^3y^2 + 4x^3y^3 + 9477x^4 + 18x^4y + 5x^4y^2 + 486x^5 + 30x^5y + 27x^6$ $L = 0.00136176272\ldots$ (open) $M_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 7 & -12 \\ 3 & -5 \end{pmatrix}$, $M_{s_1} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$	I_1 I_3 I_5 III 0 $-\frac{1}{64}$ $\frac{3}{8000}$ ∞	3^5 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(3^5) & 1 \end{pmatrix}$	
$234\underline{4}III$ $\left(\overset{\text{isog.}}{\simeq} 1\underline{2}6III\right)$	$g_2 = 1024t^3 + 144t^2 + 11/4t + 1/768$ $g_3 = -4096t^4 - 1024/3t^3 - 10t^2 - 1/24t + 1/110592$ $3\theta^2 - t \cdot (112\theta^2 + 112\theta + 36)$ $+t^2 \cdot 1024\,(\theta + \frac{5}{4})\,(\theta + \frac{3}{4})$ $a_n = 1, 12, 180, 2928, 49860, 875952, \ldots$ $F = \frac{(xy-x+3y+1)(xy+3x-y+1)}{xy}$  $L = 0.20598980413\cdots = \frac{3}{2^4}\log 3$ $M_{s_1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 4 & -3 \\ 3 & -2 \end{pmatrix}$, $M_0 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$	I_2 I_3 I_4 III $\frac{3}{64}$ $\frac{1}{16}$ 0 ∞	3^2 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(3^2) & 4 \end{pmatrix}$	$\Gamma_0(2)$

Table 10 – continued

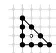
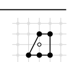
$2\underline{3}4III$ $\left(\overset{\text{isog.}}{\simeq} 12\underline{6}III\right)$	$g_2 = 64t^3 + 12t^2 - 2t + 1/12$ $g_3 = -128t^4 + 56/3t^3 - 2t^2 + 1/6t - 1/216$ $\theta^2 - t \cdot (20\theta^2 + 20\theta + 6) + t^2 \cdot 64\left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 6, 54, 588, 7110, 91476, 1224636, \ldots$ $F = \frac{(x+y-1)(2x^2-4xy+x+2y^2+y-1)}{xy}$  $L = 0.23104906019\cdots = \frac{1}{3}\log 2$ $M_{s_1} = \left(\begin{smallmatrix}1 & -2\\ 0 & 1\end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix}1 & 0\\ 3 & 1\end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix}5 & -4\\ 4 & 3\end{smallmatrix}\right)$	$I_2\ I_3\ I_4\ III$ $\frac{1}{16}\ 0\ \frac{1}{4}\ \infty$	2^2 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(2^2) & 3\end{pmatrix}$	
$\underline{2}34III$ $\left(\overset{\text{isog.}}{\simeq} \underline{1}26III\right)$	Substitute $t \mapsto t + 3/64$ $3\theta^2 - t \cdot (128\theta^2 + 128\theta + 36)$ $-t^2 \cdot 4096\left(\theta + \frac{5}{4}\right)\left(\theta + \frac{3}{4}\right)$ $a_n = 1, 12, 612, 25392, 1298340, 67041072, \ldots$ F : does not exist $L = 0.04872722313\ldots$ (open) $M_0 = \left(\begin{smallmatrix}1 & 0\\ 2 & 1\end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix}1 & -3\\ 0 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}5 & -4\\ 4 & -3\end{pmatrix}$	$I_2\ I_3\ I_4\ III$ $0\ \frac{1}{64}\ -\frac{3}{64}\ \infty$	3^4 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(3^4) & 2\end{pmatrix}$	
$11\underline{6}IV$ $\left(\overset{\text{isog.}}{\simeq} \underline{2}33IV\right)$	$g_2 = -6t^2 + 1/12$ $g_3 = -9t^4 + 1/2t^2 - 1/216$ $\theta^2 - t^2 \cdot 81\left(\theta + \frac{4}{3}\right)\left(\theta + \frac{2}{3}\right)$ $a_n = 1, 0, 18, 0, 810, 0, 45360, 0, 2806650, \ldots$ $F = \frac{(1+x)(1+x+3xy^2)}{xy}$  L : does not exist (HG pullback) $M_{s_1} = \left(\begin{smallmatrix}1 & -1\\ 0 & 1\end{smallmatrix}\right)$, $M_{s_2} = \begin{pmatrix}4 & -1\\ 9 & -2\end{pmatrix}$, $M_0 = \left(\begin{smallmatrix}1 & 0\\ 6 & 1\end{smallmatrix}\right)$	$I_1\ I_1\ I_6\ IV$ $\frac{1}{9}\ -\frac{1}{9}\ 0\ \infty$	3^3 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(3^3) & 6\end{pmatrix}$	$\Gamma_1(3)$

Table 10 – continued

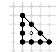
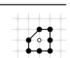
$1\underline{1}6IV$ $\left(\overset{\text{isog.}}{\simeq} 23\underline{3}IV\right)$	Substitute $t \mapsto -3t + 1/9$ $2\theta^2 - t \cdot (81\theta^2 + 81\theta + 24) + t^2 \cdot 729 \left(\theta + \frac{4}{3}\right)\left(\theta + \frac{2}{3}\right)$ $a_n = 1, 12, 198, 3720, 75690, 1626912, \ldots$ $F = \frac{(x-2y-1)(x+y+2)(2x-y+1)}{xy}$ (as for $23\underline{3}IV$)  L : see $23\underline{3}IV$ $M_{s_1} = \begin{pmatrix} -17 & -81 \\ 4 & 19 \end{pmatrix}$, $M_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 1 & -6 \\ 0 & 1 \end{pmatrix}$	$I_1\ I_1\ I_6\ IV$ $\frac{2}{27}\ 0\ \frac{1}{27}\ \infty$	2 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2) & 1 \end{pmatrix}$	
$\underline{1}16IV$ $\left(\overset{\text{isog.}}{\simeq} 2\underline{3}3IV\right)$	Same as for $1\underline{1}6IV$	$I_1\ I_1\ I_6\ IV$ $0\ \frac{2}{27}\ \frac{1}{27}\ \infty$	Same as for $1\underline{1}6IV$	
$12\underline{5}IV$	$g_2 = 10/3t^2 - 4/3t + 1/12$ $g_3 = -t^4 + 28/27t^3 - 13/18t^2 + 1/9t - 1/216$ $2\theta^2 - t \cdot (29\theta^2 + 29\theta + 8) + t^2 \cdot 27 \left(\theta + \frac{4}{3}\right)\left(\theta + \frac{2}{3}\right)$ $a_n = 1, 4, 30, 280, 2890, 31584, 358176, \ldots$ $F = \frac{(1+x)(1+x+2y+2xy+xy^2)}{xy}$  $L = 0.27725887222\cdots = \frac{2}{5}\log 2$ $M_{s_1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $M_{s_2} = \begin{pmatrix} 5 & -2 \\ 8 & -3 \end{pmatrix}$, $M_0 = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$	$I_1\ I_2\ I_5\ IV$ $\frac{2}{27}\ 1\ 0\ \infty$	2 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2) & 5 \end{pmatrix}$	$SL_2(\mathbb{Z})$

Table 10 – continued

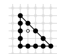
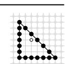
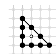
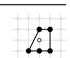
$1\underline{2}5IV$	Substitute $t \mapsto -25t+1$ $\theta^2 - t\cdot(52\theta^2+52\theta+16)+t^2\cdot 675\left(\theta+\frac{4}{3}\right)\left(\theta+\frac{2}{3}\right)$ $a_n = 1, 16, 330, 7360, 170650, 4050816, \ldots$ $F = \frac{N}{xy^2}$, where  $N = 1+y-4y^2+y^3+y^4+2x+4xy+16xy^2 +3xy^3-x^2+2x^2y-6x^2y^2-2x^3-3x^3y +x^4$ $L = 0.21520447048\cdots = \frac{1}{\sqrt{5}}\log\left(\frac{1+\sqrt{5}}{2}\right) = \frac{1}{2}L(1,\chi_5)$ $M_{s_1} = \left(\begin{smallmatrix}-1 & -4\\ 1 & 3\end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix}1 & 0\\ 2 & 1\end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix}1 & -5\\ 0 & 1\end{smallmatrix}\right)$	$I_1\ I_2\ I_5\ IV$ $\frac{1}{27}\ 0\ \frac{1}{25}\ \infty$	-1 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(-1) & 2\end{pmatrix}$	
$\underline{1}25IV$	Substitute $t \mapsto -25t+2/27$ $2\theta^2 - t\cdot(621\theta^2+621\theta+168)$ $-t^2\cdot 18225\left(\theta+\frac{4}{3}\right)\left(\theta+\frac{2}{3}\right)$ $a_n = 1, 84, 16830, 3971640, 1030948650, 282553385184, \ldots$ $F = \frac{N}{x^2y^3}$, where  $N = -1024-768y+240y^2+400y^3+60y^4-48y^5 -16y^6+768x+96xy-72xy^2-6xy^3-3xy^4 -18xy^5-960x^2-288x^2y+27x^2y^2+84x^2y^3 -9x^2y^4+400x^3-66x^3y+6x^3y^2-14x^3y^3 -240x^4-12x^4y-9x^4y^2+48x^5-18x^5y -16x^6$ $L = 0.01058254145\ldots$ (open) $M_0 = \left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix}5 & -8\\ 2 & -3\end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix}1 & -5\\ 0 & 1\end{smallmatrix}\right)$	$I_1\ I_2\ I_5\ IV$ $0\ -\frac{1}{27}\ \frac{2}{675}\ \infty$	2^5 $T = \begin{pmatrix}1 & 0\\ -\frac{1}{2\pi i}\log(2^5) & 1\end{pmatrix}$	

Table 10 – continued

$23\underline{3}IV$ $\left(\overset{\text{isog.}}{\simeq} 1\underline{1}6IV\right)$	$g_2 = 54t^2 - 4t + 1/12$ $g_3 = -729t^4 + 108t^3 - 17/2t^2 + 1/3t - 1/216$ $2\theta^2 - t \cdot (81\theta^2 + 81\theta + 24) + t^2 \cdot 729 \left(\theta + \frac{4}{3}\right)\left(\theta + \frac{2}{3}\right)$ $a_n = 1, 12, 198, 3720, 75690, 1626912, \ldots$ $F = \frac{(x-2y-1)(x+y+2)(2x-y+1)}{xy}$  $L = 0.15403270679\cdots = \frac{2}{3^2}\log 2$ $M_{s_1} = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 4 & -3 \\ 3 & -2 \end{smallmatrix}\right)$, $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right)$	$I_2\ I_3\ I_3\ IV$ $\frac{1}{27}\ \frac{2}{27}\ 0\ \infty$	2^3 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(2^3) & 3 \end{pmatrix}$	$\Gamma_1(3)$
$2\underline{3}3IV$ $\left(\overset{\text{isog.}}{\simeq} \underline{1}16IV\right)$	Same as for $23\underline{3}IV$	$I_2\ I_3\ I_3\ IV$ $\frac{1}{27}\ 0\ \frac{2}{27}\ \infty$	Same as for $23\underline{3}IV$	
$\underline{2}33IV$ $\left(\overset{\text{isog.}}{\simeq} 11\underline{6}IV\right)$	Substitute $t \mapsto -(1/3)t + 1/27$ $\theta^2 - t^2 \cdot 81 \left(\theta + \frac{4}{3}\right)\left(\theta + \frac{2}{3}\right)$ $a_n = 1, 0, 18, 0, 810, 0, 45360, 0, 2806650, \ldots$ $F = \frac{(1+x)(1+x+3xy^2)}{xy}$ (as for $11\underline{6}IV$)  L : does not exist (HG pullback) $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, $M_{s_1} = \left(\begin{smallmatrix} 1 & -3 \\ 0 & 1 \end{smallmatrix}\right)$, $M_{s_2} = \left(\begin{smallmatrix} 4 & -3 \\ 3 & -2 \end{smallmatrix}\right)$	$I_2\ I_3\ I_3\ IV$ $0\ \frac{2}{27}\ \frac{2}{27}\ \infty$	3 $T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2\pi i}\log(3) & 2 \end{pmatrix}$	

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